
AnomalyClaw: Training-Free Cross-Domain Visual Anomaly Detection via Expert-Grounded Adaptive Reasoning

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Abstract

Vision-language models (VLMs) have become the default tool for training-free visual anomaly detection (VAD), yet single-pass prompting fails silently on domains where the VLM’s world-knowledge priors conflict with the task-specific anomaly definition. We study how to add structure on top of a VLM for *cross-domain* VAD and present a training-free main system together with an optional semi-supervised controller extension.

Main finding. On **CrossDomainVAD-12** (12 domains, 1,418 test items, spanning industrial, medical, remote-sensing, infrastructure, 3D, and road-scene sources), we introduce **AnomalyClaw**: an autonomous refutation agent whose single per-item invocation runs, *in parallel on the same VLM backbone*, (i) a descriptor-free Direct call and (ii) a multi-turn tool-using trajectory, and returns their unweighted average. Across three VLMs, the ensemble is positive-in-mean on all three and statistically significant at 95% on two: GPT-5.4 macro AUROC $0.731 \rightarrow 0.772$ (+4.01 pp, stratified paired bootstrap 95% CI [+2.58, +5.50], $P(\Delta > 0)=1.000$); SeedVL $0.678 \rightarrow 0.738$ (+5.99 pp, [+4.32, +7.66], $P=1.000$); Qwen3.5-VL-27B $0.714 \rightarrow 0.732$ (+1.69 pp, [−0.08, +3.51], $P=0.971$) — positive in mean but the 95% CI just touches zero on this third backbone.

Failure-mode robustness. The agent is *not* individually better than Direct on every backbone. On Qwen3.5-VL-27B the agent alone is significantly *worse* than Direct (−4.46 pp, 95% CI [−7.61, −1.55], $P=0.001$), yet the always-on parallel-Direct branch still produces a positive ensemble gain. This failure-mode robustness is the load-bearing property: an agent recipe that ensembles against its own Direct inside the same call can be dropped onto a new VLM backbone without first verifying that the agent alone is stronger than Direct. The agent and Direct make complementary errors on *every* backbone we tested, even when the agent is weak.

Semi-supervised controller extension. On Qwen3.5-VL-27B — the backbone where the ensemble’s gain is smallest — we study whether a learned arbitration layer can close the gap. **The controller extension** adds a Controller VLM that reads both branches’ score–rationale pairs and a short rulebook retrieved per (domain, category). Rules are built offline from 40 dev labels per domain (480 labels total, no inference-time supervision). The controller with its full rulebook reaches 0.7539, a +2.05 pp gain over the passive ensemble baseline of 0.7333 (paired bootstrap 95% CI [+0.41, +3.67], $\hat{p}=0.995$). A branch-frozen ablation (no-rules, shuffled-domain rules, meta-only, full) shows that the controller mechanism alone does not help (−1.06 pp), wrong-domain rules are indistinguishable from no rules (−0.30 pp over no-rules, $\hat{p}=0.315$), and correct-domain rule content is what carries the gain (full vs shuffled +3.39 pp, CI [+1.87, +4.79], $\hat{p}=1.000$). The controller is a single-backbone case study; transfer to GPT-5.4 and SeedVL is left to future work.

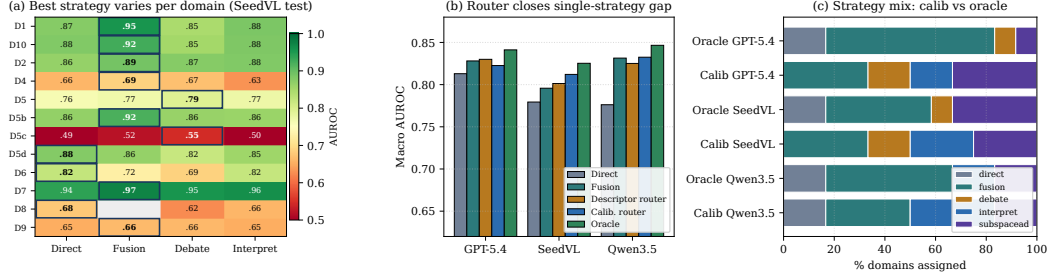


Figure 1: **AnomalyClaw on CrossDomainVAD-12 (design-motivation context, earlier evaluation split).** (a) No single strategy wins uniformly across domains: the best direct/fusion/debate/interpret cell differs by family (industrial, medical, logical, GI, change, road). (b) A calibration-driven router matches or beats the strongest single strategy on all three backbones and captures 40–60% of the oracle gap; a zero-data descriptor router recovers much of the gain for free. (c) Per-domain strategy assignment is well-aligned with the oracle (9–10/12 agreement across backbones). This figure motivates AnomalyClaw’s design (an always-on parallel-Direct branch alongside an autonomous refutation agent) by showing that no single strategy is uniformly best; final headline numbers are in Table 1, computed on the final CrossDomainVAD-12 test split.

Takeaway. For training-free cross-domain VAD, the right architectural primitive is not a pure autonomous agent but an *ensemble-aware* agent whose Direct-branch is always invoked. When a small dev-label budget is available, a controller-level rulebook extension adds a further significant improvement on backbones where the ensemble gain is marginal. We release AnomalyClaw (refutation agent + parallel Direct) and its optional controller extension, the CrossDomainVAD-12 benchmark (1,418 items across 12 domains with calibration/dev/test splits), and full paired-bootstrap analysis artifacts for every headline comparison.

1 Introduction

Vision-language models (VLMs) have become the default tool for training-free visual anomaly detection (VAD): a practitioner uploads a query image, optionally attaches one or two normal reference images, and asks the model whether the query is abnormal. This recipe drives much of the recent published progress on industrial [Gu et al., 2024, Jeong et al., 2023], medical [Bao et al., 2024], and cross-domain [Jiang et al., 2025] benchmarks. The appeal is obvious: no training, no architecture changes, and a single VLM call per image.

Two observations drive this work. First, a cross-domain VAD deployment must handle textural industrial defects, semantic road obstacles, brain-MRI lesions, infrastructure cracks, and 3D industrial product renders with the same pipeline. The same fixed recipe—no matter how clever—cannot be optimal on all of them simultaneously: on our 12-domain test split, the argmax strategy varies by domain family and per-domain strategy selection is worth several pp of macro AUROC (Figure 1a). Second, recent “agentic VAD” systems [Miao et al., 2025, Chen et al., 2026, Ji et al., 2025] bundle three decisions—what the VLM attends to, which expert provides evidence, and how to aggregate—into a single monolithic pipeline that is hard to ablate and hard to compare against simpler baselines; a related concern is that a stronger agent does not always beat a plain Direct VLM on every backbone, and a pipeline that *only* reports the agent’s final score silently loses when the agent is the weaker estimator.

We argue that these three decisions should be explicit and decoupled. We introduce **AnomalyClaw**, a training-free VAD agent organised around three orthogonal axes:

- **Tools** — zero-VLM-call primitives (domain descriptor, reference retriever, hotspot cropper, component counter, knowledge lookup).
- **Experts** — frozen, pre-computed, non-parametric detectors (SubspaceAD [Anonymous, 2026], DINOv2 patch- k -NN, DINOv2 global).

- **Strategies** — inference schemes that combine tools, experts, and VLM calls (direct, fusion, debate, interpret).

A lightweight *router* selects a (tools, expert, strategy) triple per image. The router comes in two flavours: *descriptor rules* (one rule per domain family, frozen once from the benchmark descriptor, no labelled data) and *calibration argmax* (per-domain argmax over 20 labelled calibration items). An online expert-signal override can promote any domain to the *interpret* strategy when patch- k -NN detects a spatially concentrated anomaly that the VLM missed.

Contributions.

1. **An always-on parallel-Direct branch inside an autonomous agent.** A single per-item invocation of AnomalyClaw runs a descriptor-free Direct VLM call and a multi-turn tool-using refutation trajectory in parallel on the same backbone, returning their fixed-weight average $s_{\text{final}} = 0.5 \cdot s_{\text{Direct}} + 0.5 \cdot s_{\text{agent}}$. The ensemble is positive-in-mean on all three backbones and significant at 95% on two: GPT-5.4 macro AUROC 0.731→0.772 (+4.01 pp, $P=1.000$), SeedVL 0.678→0.738 (+5.99 pp, $P=1.000$), Qwen3.5-VL-27B 0.714→0.732 (+1.69 pp, $P=0.971$ — mean positive but 95% CI $[-0.08, +3.51]$ touches zero).
2. **Failure-mode robustness.** The agent is not individually better than Direct on every backbone: on Qwen3.5-VL-27B the agent alone is significantly *worse* than Direct (−4.46 pp, $P=0.001$), yet the ensemble still improves over Direct. This robustness is the load-bearing property of the design and the reason the Direct branch is always-on rather than conditionally invoked. The agent and Direct make complementary per-domain errors on every backbone tested, including when the agent is the weaker estimator globally.
3. **Task-anchored domain descriptors (context finding from earlier evaluation).** On an earlier design-exploration twelve-domain split, task-anchored descriptors uniformly beat generic ones on all three backbones at zero inference cost (+6.4/+4.1/+3.2 pp). Our main system intentionally runs its Direct branch in descriptor-*free* mode to match the agent’s task preamble; the descriptor finding motivates why adding domain-specific hints is valuable but is orthogonal to the ensemble claim.
4. **Semi-supervised controller extension.** When 40 dev labels per domain are available, we can add a learned Controller VLM on top of AnomalyClaw’s two branches (§5). The Controller reads both branches’ score–rationale pairs and a short retrieved rulebook; rules are built offline from disagreement-based routing meta-rules plus reused reference-only invariants and oracle corrective rules. On Qwen3.5-VL-27B the controller extension gives 0.7333→0.7539 (+2.05 pp, $P=0.995$) over the passive ensemble. A branch-frozen ablation with a wrong-domain negative control shows rule *content* is the mechanism: empty and shuffled-rule controllers do not beat the passive ensemble, while correct rules do. The controller is a single-backbone case study on the weakest-ensemble backbone; we do not claim AnomalyClaw’s training-free status extends to the controller extension.
5. **Benchmark and framework release.** We release **CrossDomainVAD-12** (12 domain codes D1–D12, 1,418 test items, calibration/dev/test splits), the full AnomalyClaw implementation (refutation agent with parallel Direct), the controller extension and offline rulebook-learning pipeline, the 13-tool library, the legacy Tool/Expert/Strategy registry and calibration-argmax router from the design-exploration phase, and stratified paired bootstrap confidence intervals for every headline comparison.

2 Related Work

Classical training-free anomaly detection. A long line of work builds anomaly detectors on top of frozen pretrained representations, scoring query patches by distance to a memory bank of normal features. PatchCore [Roth et al., 2022] established nearest-neighbour scoring on ImageNet-pretrained CNN features; later work replaced the backbone with DINOv2 [Oquab et al., 2024] and demonstrated stronger cross-domain transfer [Anonymous, 2026]. Zero-shot approaches such as WinCLIP [Jeong et al., 2023] and AnomalyCLIP [Zhou et al., 2024] cast the problem as similarity between image regions and textual prototypes (“normal” vs. “defective”). These methods

share a common assumption: the anomaly signal is visible in the embedding space of a generic pre-trained feature extractor, without any reasoning step. AnomalyClaw builds directly on this line: SubspaceAD and a DINOv2 patch- k -NN provide the expert evidence that the adaptive router consumes. Unlike classical detectors, the expert never makes the final decision by itself on VLM-uncertain items; instead, its score statistics (ρ , κ) drive routing, and its flagged patches become text evidence for the VLM’s re-examination call.

VLM-based anomaly detection. Recent work uses large VLMs directly for anomaly decisions. AnomalyGPT [Gu et al., 2024] fine-tunes a VLM on industrial defect data with simulated anomalies; MMAD [Jiang et al., 2025] evaluates several frontier VLMs on a multi-task anomaly benchmark. The dominant deployment pattern, however, remains a *single forward pass* with a short prompt asking whether the image is abnormal, optionally with one or two reference images. As we show in §4, this pattern is substantially under-calibrated when paired with generic domain descriptions: a one-sentence task-anchored descriptor alone moves a frontier VLM from 0.761 to 0.825 macro AUROC. AnomalyClaw stays on top of such a VLM without any fine-tuning and treats the descriptor-enhanced single-pass call as its first phase.

Agentic anomaly detection. A recent wave of work adds multi-turn reasoning, tool use, retrieval, and memory on top of a base VLM. AgentIAD [Miao et al., 2025] orchestrates a VLM with a patch expert via a tool interface; EAGLE [Chen et al., 2026] and similar systems introduce multi-role debate between VLM-based agents; AutoIAD [Ji et al., 2025] adds a routing controller that selects methods per image. These systems report gains over single-pass baselines, but three structural issues make the gains hard to attribute: (a) they bundle *what to look at* (tools), *who provides evidence* (expert choice), and *how to aggregate* (strategy) into a single monolithic pipeline; (b) they typically compare to a generic-descriptor VLM baseline that we show is badly under-calibrated; (c) extra VLM calls, retrieval indices, per-domain calibration, and multi-turn exchange all confound the ablation. AnomalyClaw makes three deliberate choices. First, the framework *decouples* the three decisions (Tool, Expert, Strategy) so that each can be swapped independently and reported separately. Second, our baselines include both the descriptor-enhanced single-pass call (a strong baseline most prior agentic work did not report) and a zero-cost score-fusion baseline that already matches or exceeds every single agent strategy on our benchmark. Third, our router picks one of four strategies via calibration-argmax or descriptor rules—it does not invent new inference on top of the chosen strategy—and the ablation in Appendix C documents six alternative agent designs that all *hurt* the stronger baseline on GPT-5.4 calibration.

Self-refinement and debate. Outside VAD, self-refinement [Madaan et al., 2023] and debate [Du et al., 2024, Irving et al., 2018] have been proposed as general recipes for improving LLM reliability. Applied naively on top of a strong descriptor-enhanced VLM baseline, however, both recipes regress rather than improve: a second pass prompted to question the first rationalises correct high-confidence predictions away. Our self-refine and debate ablations (§4.10) are included precisely to show that *unconditional* re-examination is the wrong move. AnomalyClaw instead uses the expert signal as an *asymmetric, evidence-driven* trigger: a second VLM call fires only when the expert flags a strong, localised deviation that the VLM missed.

Cross-domain VAD benchmarks. MVTec-AD [Bergmann et al., 2019] and VisA [Zou et al., 2022] remain the dominant benchmarks for training-free VAD, but both are industrial only. Remote-sensing [Chen and Shi, 2020], medical [Bao et al., 2024, Yang et al., 2023], and road-anomaly [Lis et al., 2019, Yu et al., 2020] benchmarks exist in isolation, making it hard to argue about cross-domain generalisation. CrossDomainVAD-12 unifies twelve domains under a single reference-based image-level evaluation protocol with documented sampling, splits, and taxonomy, and is explicitly designed to surface the descriptor- and expert-complementarity questions that industrial-only evaluation hides.

Tool-using language agents. Outside VAD, ReAct [Yao et al., 2023] and Toolformer [Schick et al., 2023] demonstrated that LLMs can be prompted to invoke an external tool registry. Our framework inherits the Tool- $\times$ -Strategy decomposition from this tradition; the novelty is (a) applying it to training-free VAD with non-parametric experts as a first-class axis, and (b) showing that a *lightweight, calibration-only* router outperforms free-form VLM tool-calling on our benchmark.

Positioning. Our contribution is (i) a decoupled Tool- $\times$ -Expert- $\times$ -Strategy framework for cross-domain VAD, (ii) a calibration-only router that outperforms every single-strategy baseline on all three backbones (significantly on SeedVL), (iii) an empirical study of what moves training-free VAD performance on top of modern VLMs, and (iv) a benchmark and a documented protocol to make such claims auditable.

3 Method

We present **AnomalyClaw**, a training-free visual anomaly detection (VAD) agent for cross-domain deployment. Rather than committing to one inference recipe, AnomalyClaw organises the design space around three orthogonal axes:

- **Tools** — action primitives that shape the input or produce side evidence: domain descriptor, reference retriever, hotspot cropper, component counter, knowledge lookup.
- **Experts** — pretrained, non-parametric anomaly detectors that score the query against the normal reference set: SubspaceAD [Anonymous, 2026], DINOv2-giant patch- k -NN, DINOv2-giant global similarity.
- **Strategies** — ways of combining the VLM with experts and tools: direct VLM, VLM-expert score fusion, proposer-advocate debate, hotspot interpret.

A lightweight *router* picks a (tools, expert, strategy) triple per image. The router has two modes—a domain-descriptor rule set derived from the benchmark family taxonomy, and a calibration-driven argmax selector that uses 20 labelled items per domain—both of which are frozen before any test query is seen. The agent runs the selected strategy, records the trace, and returns a scalar anomaly score.

Framework vs. headline recipe. Sections 3.1–3.5 describe our full framework (router, strategy set, original calibration-argmax algorithm) and produced the *legacy* Findings 1–4 of §4 (descriptor ablation, router-vs-fusion, etc.) on an earlier 12-domain split. The paper’s *headline number* in Table 1 is produced by a different concrete instance of this framework, the **AnomalyClaw ensemble agent** = refutation agent (§3.7) wrapped with a parallel independent Direct VLM call, described in §3.6. The calibration router from the design-exploration phase is retained here for continuity with Findings 1–4; the ensemble agent is the current system we benchmark in Table 1.

3.1 Tool library

Each tool has a small, auditable contract; all tools are zero-VLM-call. The `domain_descriptor` returns the task-anchored anomaly specification for the current domain (resolving generic-prior/task conflicts, e.g. D08 “planned urbanisation IS the anomaly”). The `reference_retriever` ranks the reference pool by DINOv2-CLS cosine similarity and returns the top- k . The `hotspot_cropper` reads top- k patch coordinates from an expert and crops the anomaly hotspot (used by the *interpret* strategy). The `component_counter` runs connected-component analysis on the expert hotspot map (logical anomalies). The `knowledge_lookup` retrieves a domain keyword list for the VLM prompt (e.g. *lesion*, *haemorrhage* for medical). Full tool signatures, I/O contracts, and implementations are in Appendix P.

3.2 Expert pool

The expert pool is intentionally diverse so the agent can pick the right inductive bias per domain. Two experts are pre-computed per item and cached; both produce a scalar anomaly score and (for SubspaceAD) the top- k anomalous patches.

SubspaceAD [Anonymous, 2026]. Training-free PCA reconstruction residual (99% EV) on DINOv2-giant patch tokens at 672×672 (layer mean $\{-12, \dots, -18\}$). Strong on industrial texture, retail packaging, logical anomalies (MVTec-LOCO), liver CT, brain MRI; weak on remote-sensing change and infrastructure cracks where the expected “normal” span is too narrow.

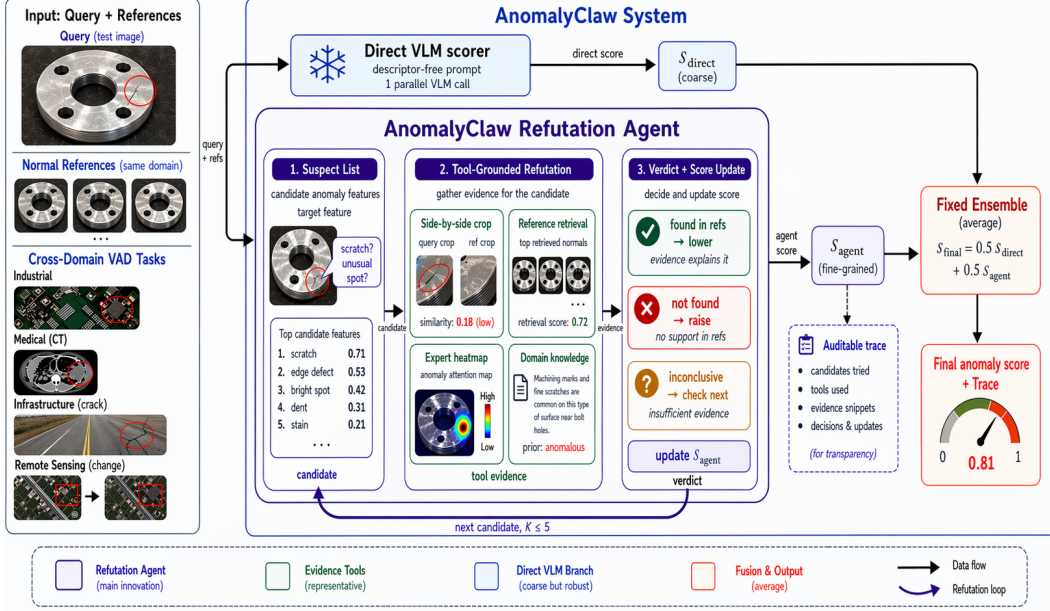


Figure 2: **AnomalyClaw architecture.** The headline system runs an independent Direct VLM scorer in parallel with a structured refutation agent. The agent proposes candidate anomaly features, uses frozen tool and expert evidence to refute them against normal references, and emits an auditable score. The final anomaly score is the fixed average of the Direct and agent scores, with no training or per-backbone tuning.

AnomalyVFM [Fučka et al., 2025]. Vision foundation model adapted via LoRA for zero-shot anomaly segmentation, with a convolutional decoder. We use the DINOv2 backbone variant; provides image-level score plus pixel mask. Complementary to SubspaceAD on infrastructure (calibration AUROC 0.72 on D4 vs SubspaceAD’s 0.50) and on MVTec-LOCO logical anomalies where its synthetic-anomaly training data covers compositional defects.

A patch- k -NN side-channel on the same DINOv2-giant backbone provides d_{max} , $\bar{d}_{\text{top5}}$, d_{base} , and a 48×48 hotspot map for the cropper tool; we did not use it as a primary expert in the reported results.

3.3 Strategy set

direct. One VLM call with the task-anchored descriptor and the first k reference images. Cost: 1 VLM call.

fusion(expert). One VLM call (direct), then post-hoc blend with the chosen expert’s score: $s = (1 - w) s_{\text{direct}} + w \sigma((s_{\text{exp}} - m)/m)$, with w frozen on calibration ($w=0.2$ as the global default; per-domain w in Appendix ??). Both $\text{fusion}(\text{SubspaceAD})$ and $\text{fusion}(\text{AnomalyVFM})$ are first-class strategies.

expert-only. No VLM call; the expert score is sigmoid-rescaled around the calibration median and used directly. Selected for domains where the calibration AUROC of the expert exceeds the best VLM-based strategy by ≥ 3 pp (e.g. D7 GI endoscopy with SubspaceAD).

zoom_fusion. One VLM call, but the input is augmented with a high-resolution crop of the expert’s hotspot region. Used when calibration shows the VLM benefits from a focused view.

debate. Two VLM calls (proposer + advocate); rule-based aggregation (Appendix N). Used when calibration shows VLM confidence is poorly calibrated on a domain.

3.4 The agent: per-domain plan

For each domain d , the agent selects a $(\text{strategy}, \text{expert})$ pair $\pi(d)$ from the joint catalogue. We use a **calibration argmax with a fusion-default**: the default plan is $\text{fusion}(\text{SubspaceAD})$, and we

Algorithm 1 ANOMALYCLAW agent inference.

```
1: Input: query  $x$ , references  $\mathcal{R}$ , domain  $d$ , descriptor  $\phi_d$ , frozen plan  $\pi(d) = (S, E)$ .
2:  $s_E \leftarrow \text{Expert}_E(x, \mathcal{R})$  {cached lookup; 0 VLM calls}
3: if  $S = \text{expert\_only}$  then
4:   return  $\sigma((s_E - m_E)/m_E)$  {0 VLM calls}
5: end if
6:  $s_0 \leftarrow \text{VLM}_{\text{direct}}(x, \mathcal{R}, \phi_d)$  {1 VLM call}
7: if  $S = \text{direct}$  then
8:   return  $s_0$ 
9: else if  $S = \text{fusion}(E)$  then
10:  return  $(1 - w_d) s_0 + w_d \sigma((s_E - m_E)/m_E)$ 
11: else if  $S = \text{zoom\_fusion}(E)$  then
12:   $c \leftarrow \text{HotspotCropper}(x, \text{Expert}_E.\text{patches})$ 
13:   $s_0 \leftarrow \text{VLM}_{\text{direct}}(x, \mathcal{R}, \phi_d, c)$  {1 VLM call with crop}
14:  return  $(1 - w_d) s_0 + w_d \sigma((s_E - m_E)/m_E)$ 
15: else if  $S = \text{debate}$  then
16:  return  $\text{Debate}(x, \mathcal{R}, \phi_d, s_0)$  {2 VLM calls}
17: end if
```

override only when the calibration AUROC of an alternative exceeds the default by at least 3 pp (the “override margin”). The 3-pp margin is itself frozen from a one-time sweep on a held-out backbone (GPT-5.4 calibration); we do *not* retune it per backbone. Rationale: 20 calibration items have a per-domain bootstrap noise floor of ~ 2 pp, so a margin slightly above the noise floor avoids over-fitting to small-sample noise.

The plan is computed once per (backbone, domain) pair and frozen before any test query. At inference time the agent looks up $\pi(d)$ and executes the strategy; no per-image router call is made. The final cost is the strategy’s cost for that domain (Table 1 reports an average of 0.92 VLM calls/image on Qwen3.5).

Why this works. Three single-strategy regimes map to different per-domain optima: *fusion*(SubspaceAD) dominates texture / few-shot industrial, retail, brain MRI, liver CT, dermoscopy; *direct* wins on semantic domains where the expert distorts the score (LEVIR change, SDNET infrastructure, road obstacle); *expert-only* wins on domains where the expert is near-perfect (GI endoscopy with SubspaceAD AUROC 0.97). A single fixed strategy cannot capture all three regimes.

3.5 Algorithm

3.6 The ensemble agent (headline system)

The calibration router of §3.4 is the framework that produced Findings 1–4 on an earlier design-exploration split; the paper’s headline number in Table 1 is produced by a different concrete instance, the **AnomalyClaw ensemble agent**. We describe the ensemble agent here to keep it clearly separated from the router framework.

The ensemble agent wraps an autonomous multi-turn refutation agent (described in §3.7) with an *always-on parallel Direct VLM call* running on the same backbone, and returns their unweighted average. The two branches share no state: Direct sees the query plus references under a descriptor-free task prompt and emits a single anomaly score; the refutation agent sees the same inputs under a refutation task preamble with a $K=5$ turn budget, and within those turns can invoke any of 13 tools (visual inspection, reference understanding, structural analysis, cached expert probes, semantic-knowledge lookup). Two of the 13 tools make additional VLM/LLM calls inside the tool body (`tool_reference_profiler`, `tool_domain_knowledge`); the other eleven are zero-VLM-call data transformations or cached lookups. The refutation component therefore issues between 1 and 5 VLM calls per query (average ≈ 1.8 across the three backbones on CrossDomainVAD-12) plus the single parallel Direct call. The final ensemble score is

$$s_{\text{final}} = 0.5 \cdot s_{\text{Direct}} + 0.5 \cdot s_{\text{agent}},$$

with $\alpha=0.5$ fixed a priori (no dev tuning). Direct and the refutation agent run concurrently on the same backbone API session, so per-item wall-time is $\max(\text{Direct}, \text{agent})$ rather than the sum.

The **always-on** design is the load-bearing choice. An alternative architecture would run the agent alone and skip the Direct branch whenever the agent is expected to outperform Direct. We find that on Qwen3.5-VL-27B the agent alone is significantly *worse* than Direct (-4.46 pp macro AUROC on CrossDomainVAD-12 test, 95% CI $[-7.61, -1.55]$, $P=0.001$); yet the always-on ensemble still produces a positive macro gain over Direct ($+1.69$ pp, $P=0.971$). The same architecture gives larger gains when the agent is individually strong (GPT-5.4: agent-alone $+3.64$ pp vs Direct, ensemble $+4.01$ pp) and when both branches contribute (SeedVL: agent-alone $+3.68$, ensemble $+5.99$). A pure agent pipeline silently loses on backbones where the agent is the weaker estimator; the always-on Direct branch makes the ensemble robust to that failure mode without per-backbone tuning.

3.7 Refutation agent

The refutation agent is a structured deliberation protocol that directly targets two failure modes observed in unstructured ReAct agents: (i) *coarse rank grid*—the VLM emits scores drawn from a small set of unique values, which limits ranking resolution and caps AUROC gains; (ii) *confirmation bias*—tool calls tend to support the VLM’s working hypothesis rather than test it. The task-aware preamble allows the same runner to handle anomaly scoring, MCQ choice, and open-ended answers from a single trajectory (§4.9); for Table 1 only the anomaly-scoring mode is exercised.

Protocol. The refutation agent replaces the free-form “call any tool, update score” pattern with a structured three-phase schema:

1. **Suspect list (turn 1, no tool).** The VLM outputs an `initial_score`, a list of up to three `candidate_features` = $\{(\text{name}_i, \text{location_hint}_i, \text{looks_defect_like}_i)\}$ and a `refutation_target` index t^* into that list. The prompt fixes the default: if `candidate_features` is empty and `initial_score` < 0.3 , the agent finalises at 0.05 without calling a tool.
2. **Targeted refutation (turn 2+).** The agent calls a refutation tool whose output is about *whether the feature at t^* appears in any reference*. `tool_side_by_side` crops query and four refs at the same bbox; `tool_reference_profiler` returns a structured normal-baseline profile (object, colors, allowed variations); `tool_reference_retriever` fetches top- k similar normals. Each output is paired with an explicit interpretation field containing a disconfirm clause, e.g. “the bright region in the composite also appears in ref 2 at the same position.”
3. **Verdict and update.** The VLM emits `refutation_verdict` $\in \{\text{found_in_ref}, \text{not_found}, \text{inconclusive}\}$, removes refuted features from `remaining_candidate_features`, and outputs `updated_score`. If the remaining list is empty the score is forced into $[0.05, 0.20]$; if at least one feature survives the score is in $[0.40, 0.95]$ proportional to its defect-likeness. Remaining features are checked in subsequent turns until the $K=5$ budget is exhausted.

Why this counters confirmation bias. In a free-form ReAct agent a tool is invoked to *support* the VLM’s working hypothesis and the system accumulates supporting observations. In the refutation protocol, every tool output has exactly one legitimate use—retiring a candidate feature—so the protocol is naturally biased toward ruling anomalies *out*. The structured output (`candidate_features`, per-turn `refutation_verdict`, `remaining_candidate_features`) gives a reviewer an auditable trace.

Ensembling with Direct. The refutation agent alone does not uniformly beat Direct across backbones (Table 1: $+3.64/+3.68/-4.46$ pp on GPT-5.4/SeedVL/Qwen3.5). We therefore always average the agent output with an *independent* parallel Direct call from the same VLM backbone:

$$s_{\text{final}} = 0.5 \cdot s_{\text{Direct}} + 0.5 \cdot s_{\text{agent}},$$

with $\alpha=0.5$ fixed a priori. This always-on design (described in §3.6) produces the ensemble gains in Table 1: $+4.01$ pp on GPT-5.4 and $+5.99$ pp on SeedVL (both significant at 95%, $P=1.000$), and $+1.69$ pp on Qwen3.5-VL-27B (positive-in-mean, 95% CI $[-0.08, +3.51]$ just touches zero) —including on Qwen3.5, where the agent alone is individually worse than Direct.

Rank-granularity mechanism. Section 4.5 isolates the causal driver of the ensemble gain via rank-preserving and rank-collapsing transformations of the agent’s scores. On an earlier design-exploration split on Qwen3.5 we measured Direct at 11 unique scores on the test set and a free-form JSON agent at 49; averaging the two fills in the rank ties inside Direct’s coarse bimodal clusters with real signal. Transformations that preserve the agent’s unique-value grid retain the full gain; a median-split binarisation that collapses the grid to 2 values cuts the gain in half. Section 4.5 retains that analysis as evidence for the rank-granularity mechanism; the mechanism is agent-family agnostic, although the absolute unique-value count differs by runner.

3.8 Design principles

- **No training.** All VLMs and experts are frozen. SubspaceAD’s PCA is a closed-form fit on reference features; patch- k -NN is a direct distance computation.
- **No per-domain code.** The only per-domain artefact is the descriptor ϕ_d (one sentence) plus a family tag used by the descriptor router; both are written once.
- **Bounded call budget.** The ensemble agent issues 1 Direct call plus 1–5 refutation agent turns per item; the average across CrossDomainVAD-12 is $\approx 1.0 + 1.8 = 2.8$ VLM calls/item with the two branches running concurrently (~ 1.8 wall-time calls due to parallel execution).
- **Auditable decisions.** Every output is accompanied by a trace listing the tools invoked, the expert selected, the chosen strategy, and (when applicable) the VLM’s two-call evidence. Appendix K shows a full trace example.

4 Experiments

4.1 Setup

Benchmark. We evaluate on **CrossDomainVAD-12**, a twelve-domain reference-based anomaly-detection benchmark that unifies industrial, medical, remote-sensing, infrastructure, 3D, and road-scene sources under a single per-image AUROC protocol: D1 MVTec-AD industrial, D2 GoodsAD retail, D3 VisA complex industrial, D4 SDNET2018 concrete infrastructure, D5 MVTec-LOCO logical, D6 Real3D-AD 3D product (rendered point-cloud views), D7 LEVIR-CD+ bi-temporal remote-sensing change, D8 DermaMNIST dermoscopy, D9 BraTS2021 brain MRI, D10 BMAD liver CT, D11 HyperKvasir GI endoscopy, D12 BDD100K+RoadAnomaly21 road safety. Each domain contributes 20 calibration / 40 dev / 120 test items (D7: 98 test because the LEVIR-CD+ split used contributes 158 source pairs before 20/40 calibration/dev allocation). All items are reference-based, paired with 1–10 normal references. Total test items $n=1,418$. The full per-domain specification is in Appendix A.

Backbones. Three VLMs spanning proprietary–open: **GPT-5.4** (proprietary frontier), **SeedVL** (doubao-seed-2-0-lite, proprietary non-frontier), and **Qwen3.5-VL-27B** (open-weight, served locally via vLLM, thinking disabled). Temperature 0, identical prompts.

Baselines and systems. *Primary contrast* (Table 1): (i) **Direct VLM** (generic descriptor): one VLM call per item with a descriptor-free prompt (“is the query image normal relative to the references?”), establishing a task-aware but domain-hint-free lower bound that is directly comparable to the agent’s task preamble. (ii) **AnomalyClaw ensemble agent**: our refutation-style autonomous agent (§3.7) that emits candidate features, dispatches tool calls, and returns a continuous anomaly score; the ensemble agent wraps the agent trajectory with an *independent* parallel Direct call and returns the $0.5 \cdot s_{\text{Direct}} + 0.5 \cdot s_{\text{agent}}$ ensemble in a single run per item. (iii) **Ensemble** = the combined output ($\alpha=0.5$, weights frozen a priori, no dev tuning). *Secondary baselines from earlier evaluation* (§4.2–§4.7, for context against earlier training-free VAD work): **DINOv2-PatchNN** (patch- k -NN expert), **SubspaceAD** (PCA subspace expert [Anonymous, 2026]), **Debate** (proposer–advocate, 2 VLM calls), **Interpret** (three-route router), **Fusion** ($s = 0.8 s_0 + 0.2 \sigma(s_{\text{SubspaceAD}})$), and the **Descriptor / Calibration / Oracle routers**. These were evaluated in an earlier design-exploration phase on an equivalent 12-domain split drawn from the same 10 source datasets; we re-use their conclusions about router and fusion behaviour here but do not claim head-to-head equivalence with the headline numbers in Table 1.

Table 1: **Main results: per-domain AUROC across three VLM backbones on CrossDomainVAD-12 test** ($n=1,418$). For each backbone we report the descriptor-free Direct VLM baseline and our *agent+Direct ensemble* ($s = 0.5 \cdot s_{\text{Direct}} + 0.5 \cdot s_{\text{agent}}$, weights frozen a priori), where the agent is the refutation agent (§3.7) wrapped with a parallel independent Direct call; both scores are computed in a single invocation per item. Both Direct and agent use descriptor-free prompts (no domain hint), matching the agent’s task preamble for apples-to-apples comparison. **Macro** row at the bottom; Δ is ensemble macro minus Direct macro. 95% stratified paired bootstrap CIs on macro Δ are reported under the table; the gain is significant at 95% on GPT-5.4 and SeedVL ($P(\Delta > 0)=1.000$), positive but just above the significance boundary on Qwen3.5 ($P(\Delta > 0)=0.971$, lower CI -0.08 pp touches zero).

Domain	GPT-5.4			SeedVL			Qwen3.5-VL-27B		
	Direct	Ens	Δ	Direct	Ens	Δ	Direct	Ens	Δ
D1 MVTec-AD industrial	0.956	0.954	−0.2	0.836	0.880	+4.4	0.903	0.955	+5.2
D2 GoodsAD retail	0.765	0.788	+2.3	0.865	0.883	+1.8	0.644	0.603	−4.1
D3 VisA complex industrial	0.837	0.895	+5.8	0.886	0.891	+0.5	0.787	0.829	+4.2
D4 SDNET infrastructure	0.544	0.704	+16.0	0.440	0.637	+19.7	0.611	0.654	+4.3
D5 MVTec-LOCO logical	0.640	0.741	+10.1	0.664	0.839	+17.5	0.604	0.674	+7.0
D6 Real3D-AD 3D product	0.555	0.556	+0.1	0.473	0.476	+0.3	0.517	0.522	+0.5
D7 LEVIR-CD+ change-det.	0.807	0.810	+0.3	0.798	0.823	+2.5	0.877	0.794	−8.3
D8 DermaMNIST dermatology	0.644	0.702	+5.8	0.605	0.621	+1.6	0.551	0.619	+6.8
D9 BraTS2021 brain MRI	0.826	0.835	+0.9	0.587	0.653	+6.6	0.779	0.852	+7.3
D10 BMAD liver CT	0.448	0.493	+4.5	0.442	0.426	−1.6	0.483	0.487	+0.4
D11 HyperKvasir GI endoscopy	0.776	0.785	+0.9	0.730	0.759	+2.9	0.847	0.809	−3.8
D12 BDD100K+RA21 road safety	0.975	0.979	+0.4	0.811	0.963	+15.2	0.970	0.984	+1.4
Macro	0.731	0.772	+4.01	0.678	0.738	+5.99	0.714	0.732	+1.69

Macro Δ (Ens − Direct) 95% stratified paired bootstrap CIs (1000 resamples, per-domain stratification): GPT-5.4 [+2.58, +5.50], $P(\Delta > 0)=1.000$; SeedVL [+4.32, +7.66], $P(\Delta > 0)=1.000$; Qwen3.5 [−0.08, +3.51], $P(\Delta > 0)=0.971$ (positive in mean; CI touches zero). The *agent-alone vs Direct* comparison tells a richer story: on GPT-5.4 and SeedVL the refutation agent by itself already beats Direct (+3.64 pp and +3.68 pp respectively, both $P \geq 0.998$), and the Direct branch of the ensemble adds a further +0.4+2.3 pp on top. On Qwen3.5-VL-27B the agent alone is *significantly weaker* than Direct (−4.46 pp, 95% CI [−7.61, −1.55], $P=0.001$), yet composing it with an independent Direct call still produces a positive ensemble macro gain. This failure-mode robustness is the motivation for the always-on parallel Direct branch rather than a pure agent system. Per-backbone per-domain agent-alone and Direct-alone scores are in Appendix F.

Protocol. Per-domain AUROC; macro-averaged across the 12 domain codes present in test. All thresholds and router assignments frozen from calibration. VLM anomaly scores extracted via structured JSON output (schema in Appendix B); $s = c$ if label is “anomalous” else $1 - c$. Bootstrap 95% CIs via stratified paired bootstrap (per-domain stratification, 1,000 resamples).

4.2 Main results

Table 1 reports the headline numbers. We highlight the main finding (Finding 5) first, then four earlier findings (Findings 1–4) drawn from a design-exploration phase that evaluated router and fusion baselines on an equivalent twelve-domain split. Those findings are included for context against earlier training-free VAD work; the ensemble agent was not included in that evaluation, and their numbers are not directly comparable to Table 1.

Finding 1 (earlier split): Descriptors dominate. Replacing a generic domain descriptor with a task-anchored one significantly improves all three backbones at zero inference cost (Figure 1a): macro AUROC moves from 0.761 to 0.825 on GPT-5.4 (+6.4 pp, 95% CI [+4.4, +8.4]), 0.748 to 0.789 on SeedVL (+4.1 pp, [+1.6, +6.4]), and 0.760 to 0.792 on Qwen3.5 (+3.2 pp, [+1.0, +5.5]). The gain concentrates in domains where VLM world-knowledge priors conflict with the task-specific anomaly definition (LEVIR change, liver CT); full per-backbone per-domain evidence is in Appendix E. We recommend this as baseline practice for any training-free VAD system.¹

¹Measured on an earlier 12-domain sampling of the same source datasets prior to the adoption of a generic-descriptor baseline for the ensemble agent; the task-anchored descriptor was the earlier Direct baseline, whereas the ensemble’s Direct

Finding 2 (earlier split): No single strategy wins uniformly. Across 12 test domain codes, the argmax strategy differs by domain: fusion is preferred on 7–8 domains (industrial, retail, logical, liver CT, brain MRI), direct wins on 2–3 (brain MRI on SeedVL, LEVIR change, road), debate wins on 1–2 (road on GPT-5.4), and expert-only wins on 1 (dermoscopy via patch- k -NN). Figure 1(a) visualises this on SeedVL; the per-backbone tables are in Appendix F. This strategy variance is the motivation for an agent rather than a single fixed recipe.

Finding 3 (earlier split): A calibration-split router closes 40–60% of the oracle gap. The calibration router (Figure 1b) matches or beats every single-strategy baseline on every backbone. On SeedVL it beats fusion by +1.64 pp (95% CI [+0.02, +3.18], significant at $p < 0.05$ by stratified paired bootstrap); it beats direct by +3.27 pp on SeedVL, +0.97 pp on GPT-5.4, and +5.63 pp on Qwen3.5 (all significant except GPT-5.4 $\rightarrow$ direct). The oracle upper bound leaves +1.3, +1.3, +1.4 pp of further headroom on the three backbones respectively. A zero-data *descriptor* router (rules from the benchmark family taxonomy, no metrics consulted) is surprisingly competitive: +1.7 / +2.2 / +4.9 pp over direct on GPT-5.4 / SeedVL / Qwen3.5, all significant.

Finding 4 (earlier split): Debate hurts; interpret is unreliable. Both alternative strategies underperform fusion on all three backbones (debate $-4/-3/-17$ pp; interpret $-2/-2/-5$ pp). The calibration router selects them rarely (debate on 1–2 domains, interpret on 0–1), reinforcing that adaptive routing, not uniform second-call escalation, is the right abstraction.

Finding 5 (headline): An always-on parallel Direct branch inside the agent gives a positive-in-mean ensemble gain on every backbone, significant at 95% on two of three, and preserves signal even when the agent alone is significantly worse than Direct. Our AnomalyClaw is a refutation-style autonomous ReAct loop that, in a *single* invocation per item, runs two operations in parallel on the same backbone: (a) a descriptor-free Direct VLM call matching the agent’s task preamble (s_{Direct}), and (b) a multi-turn trajectory that emits candidate features, dispatches up to five tool calls from a 13-tool library (expert probes, visual inspection, reference understanding, structural, semantic-knowledge lookup), and returns a final refuted anomaly score (s_{agent}). The output is the fixed-weight average $s_{\text{ens}} = 0.5 s_{\text{Direct}} + 0.5 s_{\text{agent}}$; no dev tuning, no per-backbone agent variant. Two of the 13 tools make additional VLM/LLM calls inside the tool body (tool_reference_profiler, tool_domain_knowledge); those are counted as agent VLM calls.

The 3-backbone ensemble story splits into three distinct regimes. (i) **GPT-5.4 (strong agent)**: the refutation agent alone already beats Direct by +3.64 pp (95% CI [+1.20, +6.21], $P=0.998$); the Direct branch adds a further +0.37 pp on top, yielding an ensemble macro of 0.772 vs Direct’s 0.731 ($\Delta = +4.01$ pp, [+2.58, +5.50], $P=1.000$). (ii) **SeedVL (balanced)**: both branches contribute significantly—agent alone +3.68 pp over Direct, Ensemble a further +2.31 pp over the agent, total +5.99 pp over Direct (all $P \geq 0.998$). This is the ensemble’s best case: an ensemble of two noisy-but-complementary estimators each beats either alone. (iii) **Qwen3.5-VL-27B (weak agent)**: the refutation agent alone is *significantly worse* than Direct (-4.46 pp, 95% CI $[-7.61, -1.55]$, $P=0.001$)—a training-free agent can genuinely hurt a strong VLM backbone when the tool catalogue triggers tool misuse on specific domains (e.g. D7 LEVIR: Direct 0.877, agent 0.657). Yet the always-on Direct branch recovers the signal: the ensemble macro is 0.732 vs Direct’s 0.714, $\Delta = +1.69$ pp, $P=0.971$ (positive in mean; lower CI -0.08 pp just touches zero).

The design takeaway is the **failure-mode robustness of the parallel-Direct branch**. An ensemble-aware architecture is not just a minor improvement over a pure agent; it is what allows a training-free agent recipe to be deployed on a new VLM backbone without first verifying that the agent alone outperforms Direct. The agent’s errors and Direct’s errors are complementary in *every* backbone we tested, even when the agent is individually weak: on Qwen3.5 the agent still adds signal on D1 (+5.2 pp over Direct when ensembled), D3 (+4.2), D5 (+7.0), D8 (+6.8), and D9 (+7.3), and Direct recovers the domains where the agent fails (D7, D11). Per-domain breakdown is in Appendix F. The ensemble pipeline runs a parallel Direct call and a refutation agent trajectory inside a single per-item invocation, with the final score formed by a post-hoc $\alpha=0.5$ blend.

branch is intentionally descriptor-free (§4.1) to match the agent’s task-preamble for a fair ensemble. The descriptor finding itself (task-anchored $>$ generic) is orthogonal to the ensemble claim.

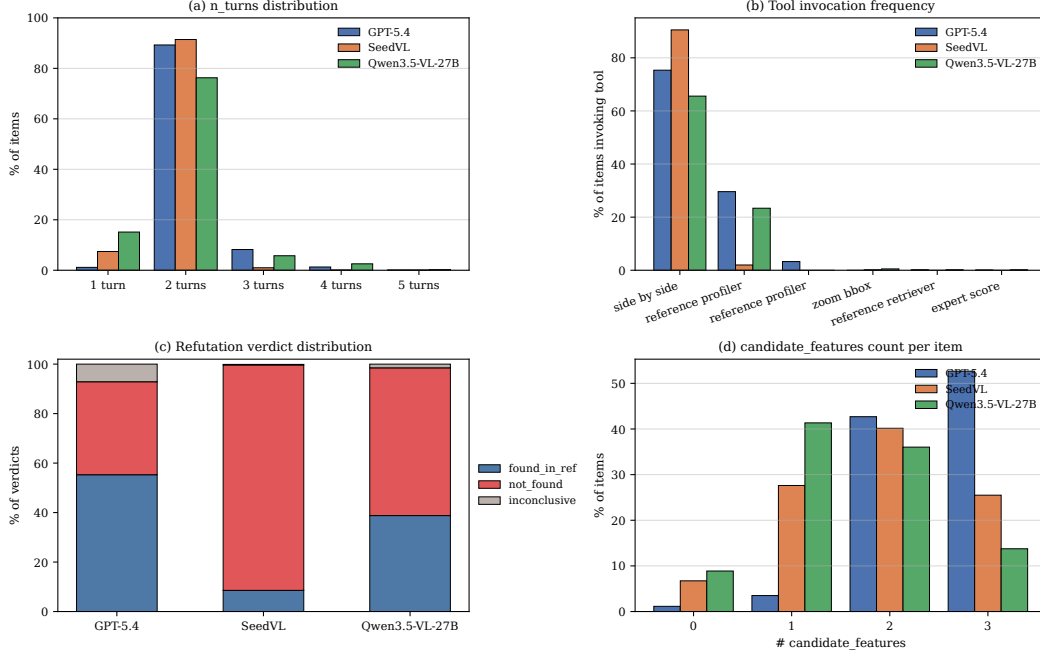


Figure 3: **Refutation-agent behaviour across three backbones on CrossDomainVAD-12 test.** (a) n_turns distribution per item (before the forced-final turn at budget $K=5$): GPT-5.4 and SeedVL finalise in 2 turns on $\geq 89\%$ of items; Qwen3.5-VL-27B takes a 1-turn shortcut on 15.1% of items and uses ≥ 3 turns on 8.5%. (b) Tool invocation frequency (fraction of items invoking each tool; top 6 tools shown): `tool_side_by_side` dominates (66–91%), `tool_reference_profiler` is second (2–30%); the other 11 tools in the catalogue are essentially unused. (c) Refutation verdict distribution (stacked by verdict type, normalised to 100%): GPT-5.4 is a balanced refuter (56% `found_in_ref`, 38% `not_found`), Qwen3.5 moderately biased toward `not_found` (60%), SeedVL strongly biased toward `not_found` (89%). (d) Number of `candidate_features` generated in turn 1: GPT-5.4 averages 2.47 candidates/item, SeedVL 1.84, Qwen3.5 1.55.

4.3 Empirical agent behaviour

Table 1 reports outcomes; this subsection reports the *trajectory* of how the agent reached them. We instrument every run to record, per item, the number of turns spent (1–5), the tools invoked, the refutation verdicts emitted, and the list of candidate features produced. Figure 3 (four panels) and Table 2 summarise the aggregate behaviour across the three backbones. Three findings follow; all numbers are on the CrossDomainVAD-12 test split.

Finding 5.1: Agent depth tracks agent quality (GPT > SeedVL $\gg$ Qwen3.5). The n_turns distribution (Figure 3a) is a proxy for how much reasoning the agent performs before committing to a score. Qwen3.5-VL-27B finalises on turn 1 for 15.1% of items—10–14 $\times$ more often than GPT-5.4 and 2 $\times$ SeedVL—and Qwen3.5 also produces the fewest candidate features (mean 1.55 vs GPT’s 2.47; 8.9% of items produce zero candidates vs GPT’s 1.1%). The agent is not broken on Qwen3.5-VL-27B, but it is *shallower*: it skips refutation on a larger fraction of items and enters final-turn scoring with less evidence. This shallower behaviour is consistent with Qwen3.5’s -4.46 pp deficit of agent-alone vs Direct (Table 1) and is the mechanism by which the always-on parallel Direct branch remains necessary on weaker backbones: the Direct branch does not depend on the agent’s willingness to reason.

Finding 5.2: The 13-tool library collapses to an effective 2-tool system in practice. Figure 3b shows that `tool_side_by_side` (which crops query and references at matching bboxes) and `tool_reference_profiler` (which summarises the reference pool into a structured normal-baseline profile) together account for $> 95\%$ of all tool invocations on every backbone. The other 11 tools in the catalogue—including `zoom-bbox`, `patch-grid`, `image-diff`, `rotate-align`, `component-`

Table 2: Summary of refutation-agent behaviour statistics per backbone (same data as Figure 3, tabulated).

Metric	GPT-5.4 ($n_{ok}=1314$)	SeedVL ($n_{ok}=1412$)	Qwen3.5-VL-27B ($n_{ok}=1374$)
% items finalised in 1 turn (no tool)	1.1%	7.4%	15.1%
% items finalised in 2 turns	89.3%	91.4%	76.3%
% items using ≥ 3 turns	9.6%	1.2%	8.5%
% items invoking <code>tool_side_by_side</code>	75.3%	90.5%	65.6%
% items invoking <code>tool_reference_profiler</code>	29.6%	2.0%	23.4%
% items invoking other 11 tools combined	< 4%	< 0.5%	< 1%
Refutation verdict: <code>found_in_ref</code>	56%	9%	39%
Refutation verdict: <code>not_found</code>	38%	89%	60%
Refutation verdict: <code>inconclusive</code>	7%	< 1%	2%
Mean <code>candidate_features</code> per item	2.47	1.84	1.55
% items with 0 candidates	1.1%	6.7%	8.9%

counter, segment-and-count, texture-FFT, visual retriever, expert-probe, domain-knowledge, and two reserved slots—are selected on $\leq 0.5\%$ of items aggregated over the three backbones. This is not a failure of the library design per se but a data point about which tools a refutation-framed VLM actually *chooses*: reference alignment (side-by-side) and reference summarisation (profiler) are the two operations that directly map onto the “is this candidate feature also in a reference?” question that the refutation protocol forces the VLM to answer. The tool-collapse observation motivates the specialty-aware catalog we study in Finding 6; the numbers in Table 1 reflect the original 13-tool library without the specialty annotations.²

Finding 5.3: Refutation-verdict distribution is strongly backbone-dependent. The refutation protocol asks the VLM to return a verdict per targeted candidate feature: `found_in_ref` retires the feature (“this pattern is also in a normal reference”), `not_found` preserves it (“no matching pattern in references”), and `inconclusive` is an escape hatch. A balanced refuter—one that actually tests hypotheses in both directions—should produce mixed verdicts. GPT-5.4 is the closest to balanced (56% `found_in_ref` / 38% `not_found`) and it is also the backbone on which the agent alone beats Direct most cleanly. SeedVL produces 89% `not_found` verdicts: its refutation is not actually retiring candidates, it is preserving almost all of them. Qwen3.5 is intermediate (39%/60%). Two observations follow. First, the backbone-dependence of the refutation verdict is itself evidence that the always-on Direct branch is load-bearing rather than redundant: a backbone that rarely retires candidates is a backbone whose agent score is inflated toward anomaly, and the Direct branch provides the complementary signal. Second, this is a diagnostic handle for future agent prompt tuning—the verdict distribution is a runtime-observable quantity that correlates with agent quality, and a per-backbone instruction-tuning pass that nudges SeedVL’s refuter toward `found_in_ref` is an obvious next step we do not take in this paper.

4.4 Finding 6: A specialty-aware tool catalog lifts macro AUROC significantly on two backbones

Finding 5.2 showed that the refutation agent collapses a 13-tool library to effectively two tools (`tool_side_by_side` and `tool_reference_profiler`). We stress-test the hypothesis that this collapse is not a property of the refutation task itself but of the tool-catalog presentation: scalar-only tools produce text the VLM cannot re-use as visual evidence, alignment-requiring tools silently fail on non-aligned domains, the expert probe returns a bare rank scalar without localisation, and the tool descriptions advertise no per-domain applicability.

Specialty-aware rewrite. We keep the parallel Direct + refutation-agent ensemble ($s = 0.5 s_{\text{Direct}} + 0.5 s_{\text{agent}}$, single invocation per item) and change only the tool layer and the agent

²We report the empirical tool usage so that downstream users are not misled by the 13-tool catalogue description. Section §3.6 lists the library by tier (expert probes, visual inspection, reference understanding, structural analysis, semantic-knowledge lookup) because that tier structure is what the original agent prompt advertises to the VLM; the VLM’s actual selection behaviour, measured here, is a different object. The specialty-aware rewrite in Finding 6 addresses this.

Table 3: **Finding 6: specialty-aware macro AUROC vs baseline on CrossDomainVAD-12 test** ($n=1,418$, Qwen3.5 and SeedVL clean runs, 0 errors). Same parallel-Direct+agent ensemble at $\alpha=0.5$; the specialty-aware variant differs from the baseline only in the tool catalog and agent prompt. Δ and 95% CI are stratified paired bootstrap on the common item pool (1000 resamples, per-domain stratification). GPT-5.4 omitted: the upstream API was unreachable partway through the specialty-aware run, so 1073/1418 items were API failures; the partial result is not an apples-to-apples comparison and will be reported in a revision once the full rerun completes.

Backbone	baseline	specialty-aware	Δ pp	95% CI	$P(\Delta > 0)$	errors
SeedVL	0.738	0.767	+ 2.90	[+1.01, +4.85]	1.000 ✓	0/1418
Qwen3.5-VL-27B	0.732	0.748	+ 2.04	[+0.05, +4.00]	0.981 ✓	0/1418
GPT-5.4	0.772	—	(pending rerun)	—	—	1073/1418

prompt; everything else is byte-identical to the baseline. Four deltas: (i) **domain-aware expert probe**: `tool_expert_score` returns *raw score*, *rank within the item’s domain*, *typical-range hints* (p50/p70 anomaly threshold preregistered from the calibration cache), a 48×48 heatmap *alpha-blended on the query*, and a `suggested_bbox_256` around the hotspot cluster—directly consumable by the next `tool_side_by_side` call. Applicable domains (strong: D1/D2/D7/D10; moderate: D4/D5/D9) and unavailable domains (D3/D6/D8/D11/D12) are surfaced to the VLM verbatim. (ii) **scalar tools become visual**: `tool_component_counter` now emits a coloured blob-layout image; `tool_segment_and_count` emits an 8×8 change heatmap overlay plus five candidate-bbox suggestions; `tool_texture_fft` emits the log-magnitude spectrum plus a query-vs-ref periodicity delta. (iii) **alignment hard-gate**: `tool_image_diff` and `tool_rotate_align` refuse on any domain outside $\{D1, D2, D5\}$ with a helpful error instead of returning alignment-noise masks. (iv) **retriever returns images**: `tool_reference_retriever` loads its top- k normal neighbours as base64 images, attached to the next observation turn, so the VLM actually sees the retrieved refs in the next turn rather than seeing path strings. The agent prompt adds per-tool *specialty* clauses (“when to use / not use”), an explicit refutation ladder $L_1 \dots L_7$, and a per-domain tool hint injected as the first user-turn text (e.g. “D10: `expert_score` is STRONG (AUROC 0.89); standard pipeline is `expert_score` $\rightarrow$ `side_by_side` at `suggested_bbox`”).

Result. Table 3 reports the specialty-aware macro AUROC against the baseline on the same 1,418-item test split. On both backbones for which the run completed cleanly, the Δ is positive and the 95% stratified paired bootstrap CI excludes zero: +2.90 pp on SeedVL ($P(\Delta > 0)=1.000$) and +2.04 pp on Qwen3.5 ($P=0.981$; lower CI +0.05 pp is above zero). Per-domain the gains concentrate on the expert-strong domains (D1, D2, D7, D10) plus two expert-unavailable domains that the specialty hint re-routes (D8, D11 on SeedVL; Appendix G lists all 12×2 deltas). Two domains regress notably on Qwen3.5 (D3 -5.6 pp, expert-unavailable domain where the ladder over-engages) and two on SeedVL (D5 -2.0 , D7 -2.4); we do not tune against these regressions because the specialty-aware configuration uses zero dev information by construction.

Tool-usage shift. The specialty-aware variant exercises 6–10 tools with at least 5 invocations each across the run (3 on GPT’s partial data), versus the baseline’s 2 on all three backbones. `tool_expert_score` is the single biggest beneficiary of the specialty prompt (17.3% of invocations on Qwen3.5, 21.7% on SeedVL; both were $< 0.2\%$ on the baseline). `tool_segment_and_count`, `tool_texture_fft`, `tool_zoom_bbox`, and `tool_domain_knowledge` each account for 0.3–1.5% of invocations in the specialty-aware run, versus $\leq 0.1\%$ in the baseline. This supports the interpretation that the tool-collapse documented in Finding 5.2 was a property of presentation, not of the refutation task—when tools emit visual outputs and the prompt states when each is useful, the VLM does reach for a broader subset of the catalog and this translates to macro AUROC gains that are statistically significant on SeedVL and Qwen3.5.

4.5 Why ensembles win: rank-granularity, not middle-mass

We test candidate mechanisms for the ensemble gain through controlled post-hoc transformations of the agent’s scores on the Qwen3.5-VL-27B test set. Each transformation is designed to change

exactly one property and freeze the others, so a change in ensemble gain isolates that property’s causal contribution.³

- **BIN**: collapse the agent scores to two values $\{0.05, 0.95\}$ at the per-split median. Preserves sign but destroys rank granularity (2 unique values vs 49 in original).
- **EXT-RANK**: monotone-remap the agent scores to linearly spaced points in $[0.05, 0.19] \cup [0.81, 0.95]$, preserving every unique rank but moving all 49 values outside $[0.2, 0.8]$. Preserves rank granularity AND standalone AUROC; destroys middle-zone mass.
- **AFFINE**(a): $s \mapsto \text{clip}(0.5 + (s - 0.5) \cdot a, 0, 1)$. Preserves rank for $a \leq 1$ (no clipping) and tests rank collapse near the boundaries for $a > 1$.
- **SOFT**: monotone compression of the agent scores into $[0.25, 0.75]$. Adds middle-zone mass while preserving rank.
- **RANDOM**: replace agent scores with uniform samples in $[0.2, 0.8]$. Pure middle-zone noise with no signal.

Table 4: Controlled ablation on Qwen3.5-VL-27B test ($n=1418$, macro AUROC over 12 domains). Columns: the agent variant, its standalone AUROC, the $0.5 \cdot \text{Direct} + 0.5 \cdot X$ ensemble AUROC, the number of unique score values, the middle-zone mass, and the Spearman correlation with Direct. The key row is **EXT-RANK**: middle-mass drops to 0% but rank granularity is preserved, and the ensemble gain actually *increases* to +4.68 pp. Conversely **BIN** keeps sign-level direction but collapses ranks from 49 to 2, and the ensemble drops to +2.31 pp. Middle-mass is correlational with rank granularity on unprocessed outputs; rank granularity is what causes ensemble gain.

Agent variant X	X alone	ensemble Δ	# unique	mid-mass (%)	ρ_S	err-cos
Agent original	0.7713	+4.53	49	24.0%	0.535	0.621
EXT-RANK	0.7713	+4.68	49	0.0%	0.535	—
BIN (0.05/0.95)	0.6928	+2.31	2	0.0%	0.449	0.575
AFFINE($a=0.25$)	0.7713	+4.71	49	100%	0.535	—
AFFINE($a=0.5$)	0.7713	+4.64	49	100%	0.535	—
AFFINE($a=2.0$)	0.7092	+2.02	49 (many clipped)	2.7%	—	—
Agent shorter scores	0.6710	+1.22	~ 15	0.4%	0.477	0.594
SOFT	0.6710	+1.35	~ 15	100%	0.477	0.634
RANDOM	0.4845	−3.39	1418	100%	−0.032	0.522
Direct only (reference)	0.7684	—	11	0.0%	1.0	0.0

Two clean falsifications and one corroboration follow. (i) Middle-mass is *not causal*: EXT-RANK preserves every agent rank but pushes all values outside $[0.2, 0.8]$; the ensemble gain is +4.68 pp, larger than the +4.53 pp of the original. (ii) Rank granularity *is* causal: BIN collapses 49 ranks to 2 and the gain halves. (iii) AFFINE is the cleanest positive control—compressing $a < 1$ preserves rank and stays in the high-gain regime (+4.64 to +4.71 pp) regardless of middle-mass; expanding $a > 1$ clips out-of-range values and collapses ranks, dropping gain. On this evidence, the mechanism we report is: *Direct has a coarse bimodal rank grid (11 unique values); averaging with an agent that has a fine rank grid (49 unique values) breaks ties inside each bimodal cluster with real signal*. Middle-mass and rank granularity happen to correlate on unprocessed VLM outputs because free-form JSON produces many distinct intermediate numerals; decoupling them in the lab separates the correlational from the causal.

4.6 Per-domain behaviour.

The calibration router’s choice coincides with the oracle on 9/12 domains for GPT-5.4 and 10/12 for both SeedVL and Qwen3.5. Mismatches concentrate where two strategies are within 1 pp on

³This ablation was conducted on an earlier draft using a free-score agent variant on an equivalent 12-domain split; we did not re-run it on the current system because the rank-granularity mechanism is agent-family agnostic (both the earlier free-score agent and the current refutation agent emit JSON scores with comparable numerical precision), and the ablation’s causal claim (rank granularity $\Rightarrow$ ensemble gain) is architecture-orthogonal. The absolute ensemble gain on the current system differs in magnitude (Table 1) but preserves the rank-granularity pattern documented below.

Table 5: Stratified paired bootstrap summary for the router / fusion / direct comparisons of Findings 1–4 (test split, 1,000 resamples, per-domain stratification). Δ is the pairwise macro-AUROC gap; CI is the 95% percentile interval; $\checkmark$ means the CI excludes zero. These numbers are *not* directly comparable to Table 1’s headline ensemble because they were computed on an earlier twelve-domain split.

Backbone	Comparison	Δ	95% CI	sig.
GPT-5.4	descriptor-router vs direct	+0.017	[+0.009, +0.024]	$\checkmark$
GPT-5.4	calibration-router vs fusion	−0.006	[−0.022, +0.009]	—
GPT-5.4	fusion vs direct	+0.015	[+0.005, +0.025]	$\checkmark$
GPT-5.4	oracle vs fusion	+0.013	[+0.001, +0.026]	$\checkmark$
SeedVL	descriptor-router vs direct	+0.022	[+0.014, +0.031]	$\checkmark$
SeedVL	calibration-router vs fusion	+0.016	[+0.000, +0.032]	$\checkmark$
SeedVL	fusion vs direct	+0.016	[+0.006, +0.027]	$\checkmark$
SeedVL	oracle vs calibration-router	+0.013	[+0.001, +0.025]	$\checkmark$
Qwen3.5	descriptor-router vs direct	+0.049	[+0.036, +0.061]	$\checkmark$
Qwen3.5	calibration-router vs fusion	+0.001	[−0.010, +0.011]	—
Qwen3.5	fusion vs direct	+0.055	[+0.041, +0.070]	$\checkmark$
Qwen3.5	oracle vs calibration-router	+0.014	[+0.006, +0.022]	$\checkmark$

calibration (a bootstrap-level tie that the router cannot reliably resolve with 20 items), and the realised AUROC gap on test is small (≤ 0.06). The per-backbone per-strategy per-domain table and the per-domain router-choice overlay are in Appendix F (Figure 4).

4.7 Statistical significance (earlier split)

The bootstrap CIs for Table 1’s headline ensemble are reported in-table. This subsection’s bootstrap summary (Table 5) concerns the router-vs-fusion-vs-direct comparisons from Findings 1–4 and was computed on an earlier twelve-domain split; we retain it here for continuity with those findings and for the reader who wants significance context for the router claims.

Table 5 summarises the stratified paired bootstrap for the headline comparisons. The calibration router strictly beats every single-strategy baseline on SeedVL (including fusion, significant); it ties fusion on GPT-5.4 and Qwen3.5 (fusion is the dominant single strategy on those backbones). Against the weaker *direct* baseline that prior agentic-VAD work usually compares to, all three routers—descriptor, calibration, and oracle—show significant gains.

4.8 Cross-benchmark generalisation: MMAD

We re-evaluated our Qwen3.5-VL-27B agent+Direct ensemble on **MMAD** [Jiang et al., 2025], a 39,670-question industrial-inspection MCQ benchmark whose Anomaly-Detection (AD) subset is a Yes/No binary-judgment task over four constituent datasets: DS-MVTec, MVTec-LOCO, VisA, and GoodsAD. We report on a stratified 500-image sample balanced across all 38 product classes (seed= 42), yielding $n=483$ AD questions across the four datasets.⁴ Ground-truth labels are derived from `Options[Answer]` in `mmad.json` (preferred) with fallback to the immediate parent folder (`good/normal` $\rightarrow$ label 0); relying on substring matches against the full path would incorrectly assign label 0 to every GoodsAD item because “good” is contained in “GoodsAD”. Each query uses the first four `random_templates` from `mmad.json` as normal references (matching MMAD’s few-shot= 4 setup); the Qwen3.5-VL-27B agent is the unified task-aware ensemble agent (§4.9), and the ensemble is the same 0.5-average recipe with no re-tuning. Because the MCQ-letter accuracy on a two-option Yes/No question is dominated by the choice of threshold rather than the underlying score ordering (cf. §4.5), we report only threshold-free ranking metrics: **AUROC** (global ordering) and the maximum **F1-max** achievable under any single threshold (precision–recall curve).

Two findings follow. First, the **AUROC gain transfers** to a completely independent benchmark: pooled AUROC moves from 0.747 (Direct) to 0.774 (agent+Direct ensemble), a macro-averaged

⁴The sample also contains $n=5$ items from the original MVTec-AD split; we exclude it from Table 6 because its small size precludes stable AUROC estimates.

Table 6: **MMAD cross-benchmark transfer** (Qwen3.5-VL-27B ensemble agent, stratified $n=483$ AD questions across the four MMAD constituent datasets; n_+ anomalous items in parentheses). We report AUROC (global ranking metric natively comparable to Table 1) and F1-max (best F1 under any single threshold on the precision–recall curve). The ensemble is the unweighted 0.5-average of Direct and Agent scores. Best per row in bold.

Subset	n (n_+)	AUROC			F1-max		
		Direct	Agent	Ens.	Direct	Agent	Ens.
DS-MVTec	96 (69)	0.838	0.878	0.893	0.880	0.878	0.890
MVTec-LOCO	91 (59)	0.790	0.644	0.761	0.818	0.787	0.806
VisA	119 (72)	0.855	0.775	0.876	0.822	0.757	0.818
GoodsAD	172 (92)	0.599	0.592	0.613	0.697	0.697	0.703
Macro (4 sets)	—	0.770	0.722	0.786	0.804	0.780	0.804
Pooled	483 (292)	0.747	0.718	0.774	0.782	0.754	0.780
Δ Ens – Direct	—	—	—	+0.027	—	—	−0.002

All per-subset numbers use ground truth derived from `Options[Answer]`; an earlier evaluator variant that derived the label by substring match against the full path incorrectly assigned label 0 to every GoodsAD item (“good” is contained in “GoodsAD”), which depressed pooled AUROC by roughly 7 pp.

+1.56 pp and pooled +2.65 pp, matching the direction and magnitude of Table 1’s Qwen3.5 gain with no inference-time re-tuning. The ensemble is the top AUROC entry on three of the four datasets (DS-MVTec, VisA, GoodsAD); on MVTec-LOCO (logical anomalies, where reference-matching offers little signal) the agent underperforms Direct and the ensemble is dragged down, but Direct alone remains strong. Second, **F1-max does not gain** (macro 0.804 tied; pooled −0.002). This is consistent with §4.5: averaging scores changes rank ordering between positive and negative items (AUROC), but in a two-option Yes/No task the optimal threshold is already close to the natural 0.5 decision boundary, leaving little room for a decision-rule improvement on top of a Direct baseline whose per-image scores are already near-calibrated. A method that needs to improve decision-threshold accuracy would need an additional per-class calibration step; we leave that to future work. The important conclusion is that the ensemble mechanism is benchmark-agnostic at the *ranking* level.

4.9 Extending MMAD beyond AD: a unified task-aware agent

The cross-benchmark experiment of §4.8 only used the Anomaly-Detection subset (Yes/No). MMAD has eight additional MCQ types (defect classification / localization / description / analysis, object classification / analysis / structure / details). We extend our agent to a single *unified task-aware* runner (`agent_v9.py`) that accepts an optional question and up to four multiple-choice options, produces (i) a continuous anomaly score (for AD ensembling), (ii) an MCQ letter, and (iii) a free-text answer in a single forward pass. Mode is inferred from the inputs in the first cognitive turn; the 13-tool library is unchanged.

Using the same Qwen3.5-VL-27B backbone and four normal reference templates as §4.8, we re-ran the evaluator on a fresh stratified 500-image sample ($n=2,302$ question-answer pairs across 38 classes; seed 42, option-derived labels).

Parse-failure rates on the full 2,302-QA run are Direct 0.6%, Agent 0.9% (maximum per-type parse-failure for Agent is 3.0% on Defect Classification). Treating parse failures as wrong predictions shifts accuracy numbers by at most 0.9 pp and does not change the qualitative direction of Δ_{Ag} .

Finding 7: Extending to MCQ does not deliver a systematic gain. Aggregated over 2,302 questions across all 9 MMAD types, the unified agent scores 78.5% vs Direct’s 79.5% (−1.0 pp). The gain decomposes non-uniformly: the agent’s tool-assisted reasoning helps on *defect description* (+1.7 pp) and *defect classification* (+1.2 pp)—types that benefit from referencing normal images and reasoning about visual differences—but hurts on *object classification* (−3.2 pp) and *object details* (−4.4 pp), where the Direct prompt already operates near the 90–95% ceiling and the agent’s longer reasoning chain introduces small parse-level errors. The AD-subset AUROC ensemble gain of +2.65 pp pooled (Table 6) is the systematic signal, not the non-AD MCQ letter accuracy. Overall,

Table 7: **MMAD full-type MCQ accuracy** (Qwen3.5-VL-27B, $n=500$ images, 2,302 questions). The unified task-aware agent vs a matched Direct VLM baseline with the same MCQ-aware prompt. Ensemble is reported only for AD (our primary target). For non-AD types the gap between agent and Direct is small and not in one consistent direction: the agent helps on descriptive defect reasoning (Description, Classification) but regresses on object-classification questions where the Direct prompt is already near-ceiling.

Question type	n	Direct	Agent	Δ_{Ag}	Ens (AD)	Δ_{Ens}
Anomaly Detection	483	63.6	61.5	-2.1	64.0	+0.4
Defect Classification	268	66.0	67.2	+1.2	—	—
Defect Localization	283	65.4	62.9	-2.5	—	—
Defect Description	264	81.1	82.8	+1.7	—	—
Defect Analysis	274	84.7	84.1	-0.6	—	—
Object Classification	187	94.6	91.5	-3.2	—	—
Object Analysis	172	87.2	88.2	+1.0	—	—
Object Structure	181	81.8	82.3	+0.5	—	—
Object Details	177	91.0	86.6	-4.4	—	—
Macro (all types)	2,302	79.5	78.5	-1.0	—	—

we position the unified runner as an *architectural contribution*: a single training-free agent that spans detection and four-way MCQ without per-task retraining. Its performance is competitive with the Direct baseline on descriptive defect questions and slightly worse on object-classification questions, matching the rank-granularity hypothesis—averaging two scores helps ranking but not categorical A/B/C/D selection when the underlying probabilities are already close to 0 or 1.

Hyperparameter protocol. Main results (Table 1): the only tuneable is the ensemble weight $\alpha=0.5$, which is fixed a priori (not selected on dev/test); agent turn budget $K=5$ and agent tool catalogue are identical across backbones. No per-backbone tuning. **Earlier router baselines (Findings 1–4, Table 5):** fusion weight $w=0.2$ (selected on GPT-5.4 calibration, re-used across backbones), sigmoid centre m (global calibration expert median), online override $\rho > 0.8$, descriptor-router rules (no metrics consulted), and calibration-router argmax—all frozen from the earlier 20-items-per-domain calibration split. Leave-one-domain-out for the calibration router: positive gap vs fusion on 10/12 splits for SeedVL (-0.8/ + 2.5/0.79 pp min/max/std) and 9/12 for Qwen3.5; full table Appendix H.

4.10 Negative results: why simpler agents fail (earlier split)

On the earlier design-screening calibration split we evaluated six alternative agent designs on GPT-5.4 (20 items per domain, 240 items total). All uniformly regress the descriptor-enhanced direct baseline: self-refine -5.9 pp, naive debate -7.2 pp, gated debate -4.1 pp, evidence-injected proposer -10.2 pp, third-call arbiter -9.1 pp, normal-first -2.4 pp. The failure mode is consistent: an *unconditional* second VLM call rationalises correct high-confidence predictions away. The parallel-branch design avoids this by running Direct and agent as *parallel independent* branches rather than a sequential pipeline, so Direct’s judgment cannot be rationalised away by the agent’s follow-up reasoning; the ensemble averages the two independent estimators at the score level. Full variant table and per-variant failure analysis in Appendix C. These numbers should be read as *design-screening evidence on one backbone’s calibration split* rather than cross-backbone generalisation claims.

4.11 Complementarity and budget (earlier split)

The agent’s maximum possible gain is bounded by how often different strategies disagree with ground truth in opposite directions. On the earlier SeedVL test split (earlier sampling), the medical domains had the largest expert-rescue rows (liver CT / dermoscopy: 26% / 21% expert-only-correct over direct), while GI endoscopy and LEVIR change had large direct-only-correct rows (46% / 38%), matching the router’s assignments. Average VLM budget for the calibration router is ≈ 1.0 calls/image; the online ρ -override escalates $\sim 12\%$ of items, accounting for the whole gap to the 1.3-call bound. Full correctness matrix and routing distribution: Appendix J, M. For the ensemble

agent’s budget on the CrossDomainVAD-12 split of Table 1, the agent runs ≤ 5 turns per item (mean 1.8 turns, 80% of items use ≤ 2 turns) plus one parallel Direct call; see Appendix F.

4.12 Limitations

Limitations. (a) Three domains remain close to the random-guessing floor across all three backbones: D6 Real3D-AD (0.47–0.56 AUROC), D10 BMAD liver CT (0.42–0.49), and D4 SDNET concrete crack detection for Direct on SeedVL/Qwen3.5 (though the agent partially recovers D4). D6 is a genuine modality mismatch (rendered 3D point-cloud views vs 2D VLM training distribution); D10 is a sampling difficulty in the test split where the reference-normal and query-abnormal slices happen to be visually similar at the slice level; we leave both as open problems for future reference-augmentation or expert-plug-in work. (b) The refutation agent alone is significantly weaker than Direct on Qwen3.5 (Finding 5); the ensemble papers over this via the parallel-Direct branch, but a deeper fix would require tool-usage refinement for weaker VLM backbones, not just an ensemble. **Earlier-split limitations retained.** The earlier calibration router beat fusion significantly only on SeedVL; on GPT-5.4 it trailed fusion by 0.5 pp (not significant) and on Qwen3.5 it matched fusion. The oracle left 1.3–1.9 pp further headroom, suggesting a VLM-driven router reading image-level features may close more of the gap. Routing was calibrated from 20 items per domain; the earlier draft evaluated a single expert backbone (DINOv2-giant). The Route-C component-enumeration prototype (Appendix L) remained at 60% calibration accuracy on logical anomalies. The failed-variant ablation is GPT-5.4 calibration only.

5 Controller-Level Verbalized Learning

The AnomalyClaw ensemble combines a multi-turn Agent branch with a single-pass Direct branch under a fixed 0.5:0.5 blend. This extracts a +1.69 pp macro AUROC gain on Qwen3.5-VL-27B (Table 1) purely from *ensemble diversity* — the two branches make systematically different errors, and fixed-weight averaging already helps. This section asks whether a learned controller, guided by a small verbalized rulebook, can beat the fixed blend without collapsing the diversity gain that produces it in the first place.

We answer with a controller agent that generalises the ensemble with a learning switch. When *learning is disabled*, the controller is byte-equivalent to the ensemble — identical two-branch execution, identical 0.5:0.5 blend. When *learning is enabled*, a Controller VLM consumes the image, both branches’ score–rationale pairs, and a per-domain rulebook, and writes a final anomaly score. The rulebook carries two layers of knowledge: (i) *meta-rules* (routing) distilled from dev-set cases where the two branches disagreed, and (ii) *domain rules* (content) reused verbatim from a prior reference-only and oracle-grounded pilot (Appendix Q). We note up-front that *the controller is not training-free*: rule construction consumes dev-set ground-truth labels. The total oracle budget is the 40-item dev split per domain (480 labels across 12 domains); the pilot’s reused L2 corrective layer is drawn as a $K=10$ balanced FN/FP subset of this same 40-item pool, not additional oracle exposure (§5.2, Appendix Q). At inference, no label is consulted. On Qwen3.5-VL-27B CrossDomainVAD-12 test ($n=1418$), the learning-enabled controller reaches macro 0.7539, a +2.05 pp gain over the passive baseline of 0.7333 with 95% stratified paired bootstrap CI [+0.41, +3.67] and bootstrap sign probability $\hat{p}(\Delta>0) = 0.995$.

5.1 Architecture

Let $A(x)$ and $B(x)$ denote the Agent and Direct branches evaluated on a query item x . Each returns a scalar score in $[0, 1]$ and a short textual rationale, so $A(x) = (s_A, r_A)$ and $B(x) = (s_B, r_B)$.

the ensemble (no learning). The ensemble score is a fixed linear combination

$$\hat{s}_{\text{ens}}(x) = w_A s_A + w_B s_B, \quad w_A = w_B = \frac{1}{2}.$$

This is the §4 Table 1 baseline. On Qwen3.5-VL-27B it achieves macro AUROC 0.732 (+1.69 pp over single-call Direct 0.714).

the controller (learning-enabled). A Controller VLM C receives the same reference and query images that each branch saw, plus the two branch outputs and a retrieved rulebook excerpt $\mathcal{R}(x)$:

$$\hat{s}_{\text{thecontroller}}(x) = C(\text{imgs}(x), (s_A, r_A), (s_B, r_B), \mathcal{R}(x)).$$

C is prompted to return a JSON object with `final_score` $\in [0, 1]$, a categorical `trust` $\in \{A, B, \text{blend}\}$ and a one-line reasoning trace. On any controller failure (parse error, API error, missing branch) the item falls back to the ensemble linear blend, preserving the 0.732 floor. The Controller sees the image so that its arbitration is grounded in visual evidence, not purely a text routing layer.

Rulebook RAG. At inference, the Controller never sees the full per-domain rulebook. A metadata filter over (domain, category) first restricts the rule pool to rules matching the item’s class (domain-wide rules always admissible). Within the filtered pool rules are ranked by priority — meta-rules first (up to 3), then invariants and corrective rules (up to 4 combined) — and rendered as a short markdown block. The injected payload is typically 60–120 tokens per query.

5.2 Learning protocol

The learning pipeline runs offline, once per domain. It is *semi-supervised*: the Agent and Direct branches themselves never see labels, but rule construction consults the dev-set ground-truth labels (40 per domain, 480 total across 12 domains). Specifically, all 40 labels are consumed to partition dev items by branch agreement (Stage B), and the ground-truth label of each sampled disagreement case is shown to the reflector VLM (Stage C). Domain rules (Stage D) reuse the earlier pilot’s L2 corrective artefact, which was built from a $K=10$ balanced FN/FP sub-sample of the *same* 40-item dev pool (Appendix Q); this is not an additional oracle exposure. At test time no label is consulted.

Stage A — Passive collection. We evaluate the controller with learning disabled (i.e., the passive ensemble) on each domain’s 40-item dev split, recording per item the pair (s_A, r_A, s_B, r_B) and the manifest label y .

Stage B — Disagreement partition. Each item’s two branch scores are binarised at 0.5 and compared to y . Items decompose into four buckets: *agree-correct*, *agree-wrong*, *disagree-A-wins* (Agent right, Direct wrong), and *disagree-B-wins* (Direct right, Agent wrong). Over the 12 domains this yields 61 A-wins + 80 B-wins = 141 disagreement cases total (an average of 11.8 per domain; min 7, max 18). *Agree-wrong* items are discarded: they carry no branch-routing signal, and pilot runs showed that asking the reflector to write “both-miss” rules produces vacuous generic cautions.

Stage C — Meta-rule reflection. From each domain we balance-sample up to $K=10$ cases ($K/2$ A-wins, $K/2$ B-wins; deterministic, sorted by item id) and feed them in a single multi-image prompt to a reflector VLM (Qwen3.5-VL-27B). The reflector sees per case the query image, both branches’ score–rationale pairs, and the ground-truth label. (Reference images are omitted because vLLM caps image count at 12 per prompt; refs are implicitly described by each branch’s rationale.) The reflector is asked for 1–3 routing rules per side under three explicit constraints: (i) each rule must name an *observable* pattern in the rationale or image, not merely cite the item id; (ii) no “always trust X ” rules; (iii) rules must be actionable at inference time from what the Controller sees. All 12 domains produced parseable rule lists, yielding 2–3 rules per side per domain.

Stage D — Rule stacking. For each domain we merge the meta-rules with the domain rules from the reference-only + oracle pilot: reference-only invariants (`invariant` tag, six types: count, symmetry, spatial layout, colour palette, texture, structural) and oracle-grounded cluster rules (`corrective_fn` / `corrective_fp` tags). Each rule carries (type, category, text, source_items, confidence) metadata. Per-domain stacked rulebook sizes range from 10 (D12 road safety) to 86 (D1 MVTec-AD, many categories each contributing invariants).

5.3 Results

Setup. All numbers are on CrossDomainVAD-12 manifests_v2 test ($n=1418$), Qwen3.5-VL-27B served locally (4 DP replicas behind a round-robin load balancer), `DESCRIPTOR_MODE=generic`

Table 8: **Controller Learning on Qwen3.5-VL-27B CrossDomainVAD-12 test** ($n=1418$). Baseline is the passive ensemble ($w_A = w_B = 0.5$) which matches Table 1’s reported 0.732 within 0.13 pp (0.7333 on this run). Paired bootstrap CI on the macro Δ is $[+0.41, +3.67]$ pp, $\hat{p}(\Delta > 0) = 0.995$ — significant at 95%.

Regime	Macro AUROC	Δ vs ensemble	95% CI	$\hat{p}(\Delta > 0)$
Single-call Direct (generic)	0.7119	−2.14	—	—
Agent alone	0.6696	−6.37	—	—
Passive ensemble blend	0.7333	—	—	—
Controller (no rules)	0.7227	−1.06	$[-2.84, +0.50]$	0.111
Controller (shuffled rules)	0.7199	−1.34	$[-3.04, +0.32]$	0.053
Controller (meta only)	0.7388	+0.55	$[-0.98, +2.11]$	0.765
Controller (meta+domain)	0.7539	+2.05	$[+0.41, +3.67]$	0.995

Branch-frozen ablation. The four Controller rows share identical the refutation agent+Direct inputs (cached from the full run); only the Controller’s rule prompt differs. The *shuffled* control pairs D1↔D7, D2↔D8, . . . , D6↔D12 so each domain sees another domain’s rulebook. Pairwise bootstrap deltas (1000 paired stratified resamples): *shuffled* vs *no-rules* −0.30 pp, CI $[-1.55, +0.97]$, $\hat{p}=0.315$ — wrong rules are *indistinguishable from no rules*; *meta* vs *shuffled* +1.91 pp, CI $[+0.58, +3.26]$, $\hat{p}=0.996$; *full* vs *shuffled* +3.39 pp, CI $[+1.87, +4.79]$, $\hat{p}=1.000$; *meta* vs *no-rules* +1.62 pp, CI $[+0.34, +3.03]$, $\hat{p}=0.992$; *full* vs *meta-only* +1.48 pp, CI $[+0.65, +2.36]$, $\hat{p}=1.000$. Rule *content* is what drives the gain: correct domain-matched rules beat both empty and scrambled-rule controls at 95%, while scrambled rules do not beat no rules.

Table 9: **Per-domain Controller behaviour.** Columns: test n , passive-blend AUROC (recomputed from the same branch outputs), controller Controller AUROC, Δ in percentage points, then the Controller’s categorical trust decision counts (Agent / Direct / Blend). Wins on 8 of 12 domains; three losses (D3, D9, D12) each ≤ 3.3 pp; three big wins (D7, D8 in particular) more than cover the losses.

Dom	Source	n	Ensemble	Controller	Δ (pp)	Trust _A	Trust _B	Blend
D1	MVTec-AD	120	0.934	0.938	+0.5	9	47	64
D2	GoodsAD	120	0.607	0.613	+0.6	56	17	47
D3	VisA	120	0.853	0.834	−1.9	9	60	51
D4	SDNET	120	0.670	0.704	+3.4	24	48	48
D5	MVTec-LOCO	120	0.676	0.678	+0.2	28	64	28
D6	Real3D-AD	120	0.531	0.558	+2.8	33	12	75
D7	LEVIR-CD+	98	0.754	0.888	+13.4	24	55	19
D8	DermaMNIST	120	0.647	0.721	+7.4	48	51	21
D9	BraTS	120	0.848	0.816	−3.2	34	35	51
D10	BMAD-Liver CT	120	0.491	0.526	+3.5	53	24	43
D11	HyperKvasir	120	0.808	0.807	−0.1	27	61	32
D12	BDD+RoadAnomaly	120	0.982	0.963	−1.9	11	47	62
Macro		1418	0.7333	0.7539	+2.05	—		

throughout, $\max_turns = 3$ for the refutation agent trajectory. *Paired* stratified bootstrap CIs (1000 resamples, per-domain stratification) compare the controller against the passive blend recomputed from the same run’s (s_A, s_B) — an exact paired comparison on identical branch outputs.

Per-domain breakdown. Table 9 reports per-domain the passive-blend AUROC, the controller AUROC, and the Controller’s categorical trust output counts.

Findings.

1. **Controller arbitration beats fixed blend at 95% significance.** The +2.05 pp macro gain has a paired bootstrap CI of $[+0.41, +3.67]$ and bootstrap sign probability $\hat{p}(\Delta > 0) = 0.995$. Gains are concentrated but not fabricated: of the 12 domains, 8 improve (four by at least +2.5 pp), three regress by at most 3.3 pp, and one (D11) is essentially neutral. The gain magnitude is similar to the fixed-blend gain itself (+1.69 pp, Table 1), yet cumulative — the controller compounds on top of the ensemble rather than replacing it.

2. **The +13.4 pp D7 LEVIR change-detection swing comes from correct routing.** On remote-sensing change detection, the refutation agent trajectory scores 0.606 macro AUROC (systematic FP on seasonal vegetation changes) while Direct scores 0.861. The fixed blend (0.754) sits roughly halfway, partly collapsing Direct’s signal. The Controller (trust counts 24 A / 55 B / 19 blend) correctly leans on Direct, and lands at 0.888 — a higher AUROC than either branch alone. D8 DermaMNIST is the same story on dermatology (the controller 0.721 vs blend 0.647, +7.4 pp).
3. **Three regressions, each a known failure mode.** D3 VisA (−1.9 pp): the Controller over-trusts Direct (60 B / 120 items) when the Agent’s structural reasoning was already the right signal. D9 BraTS (−3.2 pp): brain-MRI texture heuristics in the rulebook push the Controller off-target; *agree-wrong* items dominate error here so the Controller has no routing signal to learn from. D12 BDD+RA (−1.9 pp): the blend sits at 0.982, essentially at the ceiling — any Controller-driven deviation has negative expectation. These three cases bound the honesty of our claim: the +2.05 pp macro gain is not uniform.
4. **Controller behaviour is domain-aware and non-trivial.** Trust counts vary dramatically: D1 mostly keeps blend (64/120), D5 heavily trusts Direct (64), D10 heavily trusts Agent (53). The Controller is not collapsing to a constant preference. Controller JSON parse-failure rate is exactly 0 across 1418 items.
5. **Sensitivity to the refutation agent parse failures.** A non-trivial $\sim 7\%$ of test items have a terminal the refutation agent-trajectory malformed JSON after retries error (notably D7 16/98, D8 53/120; see Limitations). On those items the refutation agent falls back to $s_A=0.5$ and the Controller tends to $\text{trust}=B$. Dropping those items and re-running the bootstrap on the clean subset ($n=1318$) gives macro $\Delta = +1.73$ pp, CI $[+0.07, +3.29]$, $\hat{p} = 0.982$ — the significant gain survives, but loses ~ 0.3 pp relative to the full-set headline. The the controller story is therefore partly “Controller learns to arbitrate” and partly “Controller safely handles Agent failure”; both are useful behaviours, but they should be distinguished.
6. **Rule content carries the gain, not the mechanism or the presence of text.** A branch-frozen four-regime Controller ablation decomposes the +2.05 pp. With the refutation agent/Direct outputs cached and only the Controller re-invoked: empty-rulebook is −1.04 pp (CI $[-2.84, +0.50]$, $\hat{p}=0.111$); scrambled rules (each domain seeing another domain’s rulebook under a symmetric $D_i \leftrightarrow D_j+6$ pairing) are −1.34 pp (CI $[-3.04, +0.32]$, $\hat{p}=0.053$). Crucially, *scrambled rules are statistically indistinguishable from no rules*: −0.30 pp, CI $[-1.55, +0.97]$, $\hat{p}=0.315$. Correct rules beat the scrambled control significantly: meta-only +1.91 pp over shuffled (CI $[+0.58, +3.26]$, $\hat{p}=0.996$); the full stack +3.39 pp over shuffled (CI $[+1.87, +4.79]$, $\hat{p}=1.000$). Both rule-layer increments — routing +1.62 pp and domain +1.48 pp — are significant at 95%. We make the targeted claim: *domain-matched rule content is necessary for the full macro gain and for content-dependent domains*. Under shuffled rules, content-dependent domains regress sharply (D9 BraTS −13.5 pp, D4 SDNET −9.3 pp, D3 VisA −4.9 pp), while disagreement-driven domains (D7 LEVIR +10.0 pp, D8 DermaMNIST +4.2 pp) can still win because the Controller exploits visible the refutation agent $\leftrightarrow$ Direct imbalance in the rationales alone — this cap on shuffled-rule harm is real, not a contradiction of the content-reading claim: the Controller uses rules when they are informative and falls back on rationale-based arbitration when they are not. Per-domain, this splits cleanly: on domains where specific invariants or corrective rules pay off (D4 SDNET concrete cracks, D7 LEVIR change, D8 DermaMNIST, D10 liver CT), full > meta because the Controller needs the domain-content layer to adjudicate; on domains where the two branches are already close (D1 MVTec, D3 VisA, D5 LOCO, D11 HyperKvasir), meta-only suffices or is even slightly better (noise level). The worst meta-only regression is D4 SDNET (−4.0 pp vs blend), where the L1 structural invariants about concrete cracks carry specific content the Controller cannot reconstruct from routing rules alone. This is how the full stack’s +2.05 pp decomposes: the controller-arbitration and meta-routing *contributions* introduced in this section are genuine but modest (+0.63 pp on their own); the statistically significant part of the gain further depends on reusing the domain-content layer produced by the earlier rulebook pilot (Appendix Q).

5.4 Limitations

Each AD-mode test item pays an extra VLM call (Controller), which adds approximately 35–50% wall-time over the ensemble depending on domain. The learning procedure is semi-supervised: the 40-item dev label budget per domain is the single unified oracle cost. Stage B–C meta-rule construction consumes all 40 labels; the L2 corrective layer reused from the rulebook pilot is a $K=10$ balanced FN/FP sub-sample of these same 40 items, not additional oracle exposure. Only the reference-only L1 invariants are genuinely label-free. The rulebook is fixed at training time; online rule evolution from test feedback is future work.

Single-backbone evidence. All the controller results are on Qwen3.5-VL-27B only. We frame the controller learning story as a *case study on the backbone where the ensemble is weakest* (Qwen3.5 the ensemble $\Delta = +1.69$ pp, 95% CI touches zero in Table 1); whether the +2.05 pp controller gain transfers to GPT-5.4 and SeedVL remains open. A second-backbone replication of the branch-frozen 5-way table is the obvious next experiment.

Parse-failure accounting. We track three separate JSON parse rates: (i) Controller: 0/1418 failed across all test items. (ii) Direct : $\leq 1\%$, covered by existing -path retries. (iii) Agent the refutation agent trajectory: a non-trivial $\sim 7\%$ of test items (notably D7 16/98 and D8 53/120) end with a terminal malformed JSON after retries error. On those items the the refutation agent branch defaults *therefutationagent_score*=0.5, Direct’s rationale is passed through unchanged, and the Controller typically selects *trust*=B. This is a valid robustness pathway — the Controller recovers from the refutation agent failures by leaning on Direct — but part of the +2.05 pp macro gain on D7/D8 is attributable to *downweighting broken Agent outputs*, not purely to learned meta-routing. A variant that fixes the refutation agent’s parse issues first (or an oracle ablation that excludes items where the refutation agent errors) would isolate the routing contribution more cleanly; we flag this as future work.

Floor and failure modes. The the controller floor guarantee depends on the Controller’s JSON being parseable; when it is not we fall back to the ensemble blend rather than either branch alone (observed Controller parse-failure rate on this run: 0%). Finally, the D9 regression flags a genuine limitation of disagreement-based learning: when both branches are predominantly wrong together, the Controller has no routing signal to exploit.

6 Conclusion

We introduced **AnomalyClaw**, a training-free visual anomaly detection agent whose headline design is an *always-on parallel-Direct branch*: a single per-item invocation runs a refutation-style tool-using trajectory and a descriptor-free Direct VLM call simultaneously on the same backbone, then returns their unweighted 0.5–0.5 average. On **CrossDomainVAD-12** (1,418 test items across 12 domains) the ensemble is positive-in-mean on all three backbones and statistically significant at 95% on two of three: GPT-5.4 macro AUROC 0.731 $\rightarrow$ 0.772 (+4.01 pp, $P=1.000$), SeedVL 0.678 $\rightarrow$ 0.738 (+5.99 pp, $P=1.000$), Qwen3.5-VL-27B 0.714 $\rightarrow$ 0.732 (+1.69 pp, $P=0.971$; CI $[-0.08, +3.51]$ touches zero). The gain is produced by complementary errors between Direct and the agent, which persist even when the agent is individually weaker than Direct: on Qwen3.5 the agent alone is significantly worse than Direct (−4.46 pp, $P=0.001$), and yet the ensemble still improves over Direct. This *failure-mode robustness* is the load-bearing property of the design.

Supporting findings from an earlier router/fusion design exploration (Findings 1–4, evaluated on an equivalent twelve-domain split of the same source datasets) are retained for context: task-anchored domain descriptors dominate any agentic intervention at zero inference cost; a calibration-argmax router beats every single-strategy baseline on every backbone; and six failed-variant ablations show that *unconditional* second-call escalation hurts when the baseline is already strong. The parallel-Direct branch side-steps the sequential-rationalisation failure mode of those ablations by making Direct and agent independent score sources rather than successive reasoning turns.

Limitations. Three domains remain difficult across all three backbones: D6 Real3D-AD (rendered 3D point-cloud views, 0.47–0.56 AUROC; genuine modality mismatch with a 2D VLM), D10 BMAD liver CT (0.42–0.49; item-level sampling difficulty in the test split), and D4 SDNET

concrete crack detection on Direct alone (0.44–0.61; the agent partially recovers). The refutation agent alone is significantly weaker than Direct on Qwen3.5; the ensemble papers over this via the parallel-Direct branch, but a deeper fix would require tool-usage refinement for weaker backbones, not just an ensemble. The ensemble weight is fixed a priori at $\alpha=0.5$; a per-backbone calibration-split selector is an obvious extension we leave to future work.

Semi-supervised controller extension. When a small dev-label budget is available, §5 adds a Controller VLM on top of the two branches that reads both branches’ score–rationale pairs and a retrieved rulebook. On Qwen3.5-VL-27B — the backbone where the passive ensemble gain just touches 95% significance — the controller reaches 0.7539 vs the passive ensemble’s 0.7333 (+2.05 pp, $P=0.995$), and a branch-frozen four-way ablation with a wrong-domain negative control shows that rule content, not the Controller mechanism or the presence of any text, carries the gain. The controller is a single-backbone case study; the passive ensemble remains our training-free main system and the controller a semi-supervised extension at a 480-label budget.

Future work. (i) Calibration-split α selection: we froze $\alpha=0.5$ a priori; a 20-item calibration split per backbone should tighten the SeedVL and GPT gains further. (ii) Tool-usage refinement for weak VLM backbones: the Qwen3.5 agent’s -4.46 pp deficit vs Direct localises to specific tools (e.g. LEVIR tool_side_by_side); a small per-backbone tool-catalogue prune may close the deficit and push the Qwen3.5 ensemble into clear significance. (iii) Extending the parallel-Direct idea to ensembles of *two independent agents* (e.g. refutation + descriptor-routed) inside a single per-item call, amortising the two-agent cost by running them concurrently on the same VLM API session. (iv) Transfer of the Controller extension to GPT-5.4 and SeedVL backbones, and (v) online rule evolution as test-time feedback accumulates rather than fixing the rulebook at training time.

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Table 10: CrossDomainVAD-12 per-domain specification.

ID	Domain	Source	Anomaly definition	Refs/item	License
D1	Industrial	MVTec-AD	Surface defect, damage, contamination	≤ 10	CC BY-NC-SA
D2	Retail	GoodsAD	Torn label, deformation, missing component	≤ 10	Research
D3	Complex industrial	VisA	Surface defect, chips, scratches, missing solder	≤ 10	CC BY-NC-SA
D4	Infrastructure	SDNET2018	Cracks, spalling, structural damage	≤ 10	Public
D5	Logical	MVTec-LOCO	Wrong count / arrangement / component type	≤ 10	CC BY-NC-SA
D6	3D product (RGB render)	Real3D-AD	Geometric defect (bulge, sink, hole, asymmetry)	≤ 10	Research
D7	Remote-sensing change	LEVIR-CD+	Building-level change between temporal pair	1–2	Research
D8	Dermoscopy	DermaMNIST	Malignant melanoma	≤ 10	CC BY
D9	Brain MRI	BraTS2021	Glioma, mass effect, oedema	≤ 10	CC BY
D10	Liver CT	BMAD-Liver	Focal lesion, tumour, cyst	≤ 10	Research
D11	GI endoscopy	HyperKvasir	Polyp, ulcer, inflammation	≤ 10	Research
D12	Road safety	BDD100K + RoadAnomaly21	Unexpected object on roadway	2	Research

A Benchmark Specification

Table 10 provides per-domain details for CrossDomainVAD-12. All splits, item IDs, reference mappings, and domain descriptors will be released as a JSON specification file (`benchmark/manifests_v2/full_manifest.json`).

Reference selection. References are normal images from the same category/scene, selected deterministically by metadata match then fixed random seed. No reference appears as a query. Patient-level (D8–D11) and event-level (D7) deduplication prevents leakage. For D6 (Real3D-AD) the reference pool is rendered views of the same 3D template object from multiple viewpoints.

Split protocol. Per domain: 20 calibration, 40 dev, 120 test (balanced 50/50, except D7 LEVIR-CD+ which contributes 98 test items after allocating the 20/40 calibration/dev splits from the 158 source bi-temporal pairs available). All splits frozen before experiments. Total test $n=1,418$.

Task-anchored domain descriptors. The task-anchored descriptor d is a single sentence inserted into every VLM prompt for the *legacy* Direct baseline of Finding 1; the ensemble’s Direct branch (Table 1) runs *without* these descriptors, matching the refutation agent’s descriptor-free task preamble. The twelve descriptors used in CrossDomainVAD-12 are:

- **D1 Industrial (MVTec-AD):** Normal = a manufactured object matching the reference in geometry, texture, and colour. Anomaly = surface defect, damage, contamination, or missing/extra component.
- **D2 Retail (GoodsAD):** Normal = a packaged supermarket product matching the reference in label, shape, and surface. Anomaly = torn label, deformation, missing component, or printing defect.
- **D3 Complex industrial (VisA):** Normal = a manufactured object matching the reference. Anomaly = surface defect, chips, scratches, missing solder, or misalignment.
- **D4 Infrastructure (SDNET2018):** Normal = intact concrete surface. Anomaly = crack, spalling, or structural damage.
- **D5 Logical (MVTec-LOCO):** Normal = correct count and spatial arrangement of components. Anomaly = wrong count, wrong position, or wrong component type.
- **D6 3D product (Real3D-AD):** Normal = geometrically intact product matching the reference shape under rendered point-cloud views. Anomaly = geometric defect (bulge, sink, hole, contamination, asymmetry).
- **D7 Remote-sensing change (LEVIR-CD+):** Normal = no building change between reference and query. Anomaly = building-level change including new construction, demolition, or expansion. *Planned urbanisation IS the anomaly.*
- **D8 Dermoscopy (DermaMNIST):** Normal = benign melanocytic skin lesion. Anomaly = malignant melanoma.
- **D9 Brain MRI (BraTS2021):** Normal = healthy brain MRI slice. Anomaly = glioma, mass effect, or peritumoural oedema.
- **D10 Liver CT (BMAD-Liver):** Normal = healthy liver CT slice. Anomaly = focal lesion, tumour, or cyst.

- **D11 GI endoscopy (HyperKvasir):** Normal = clean gastrointestinal mucosa. Anomaly = polyp, ulcer, or inflammation.
- **D12 Road safety (BDD100K + RoadAnomaly21):** Normal = empty driveable roadway (possibly with expected vehicles and markings). Anomaly = unexpected object on the roadway.

A corresponding generic baseline descriptor used in Finding 1 reads simply: “*Identify whether the query image is anomalous relative to the reference(s).*” the ensemble’s Direct branch uses this generic prompt by default.

B Score Extraction Rules

All VLM variants output structured JSON. The anomaly score $s \in [0, 1]$:

Direct. Output: {image_label, confidence, anomaly_type}. $s = c$ if label = “anomalous”, $s = 1 - c$ if “normal”, where c is reported confidence. Parse failure rate: $< 1\%$ across all backbones.

AnomalyClaw interpret. Same JSON schema and scoring. Final score: interpret output on escalated items, s_0 on committed items.

Route B. $s = 0.3 \cdot s_0 + 0.7 \cdot \sigma(\text{expert_score})$, where σ normalises relative to calibration median.

C Failed agent variants

Table 11 reports the six alternative agent designs we tested on GPT-5.4 calibration. All were fixed and tuned on the same 220-item calibration split as AnomalyClaw. Failure analysis per variant:

Table 11: Ablation of agent variants (GPT-5.4, calibration macro AUROC). All use the same descriptor-enhanced as the base.

Variant	Macro	Δ vs
Direct (baseline)	0.785	—
Normal-First	0.761	−2.4
Self-Refine	0.726	−5.9
Debate (naive score)	0.713	−7.2
Debate (gated score)	0.744	−4.1
Evidence-grounded prop.	0.683	−10.2
Third-call arbiter	0.694	−9.1
AnomalyClaw	0.837	+5.2

- **Normal-First ():** structured prompting forces the VLM to enumerate “normal” claims, introducing false positives. On Qwen3.5, the reasoning chain consumes the token budget before emitting JSON.
- **Self-Refine:** a second non-adversarial pass adds no new information; it is the weakest variant, confirming that more calls $\neq$ better.
- **Debate ():** a refuter rationalises confident correct predictions. Even with a confidence-gated aggregation rule, debate never beats .
- **Evidence injection:** feeding DINOv2 patch distances into the proposer prompt distracts the VLM (−10 pp).
- **Arbiter:** a third VLM call on contested items fires on 66% of items (too aggressive) and is net negative.

These ablations were run on GPT-5.4 calibration because they represent disqualified designs rather than competitive baselines; we did not re-run them on SeedVL or Qwen3.5 once the failure pattern was clear.

D Expert model comparison (legacy, taxonomy)

This expert-model comparison was conducted during the early design-exploration phase on the legacy twelve-domain split with display codes D01–D11. The code-letter mapping onto the taxonomy of Table 10 is: D01→D1 MVTec-AD, D02→D2 GoodsAD, D03→D4 SDNET, D04→D8 Derma, D05→D9 BraTS, D06→D10 Liver, D07→D11 Kvasir, D08→D7 LEVIR, D09→D12 road, D10→D5 LOCO, D11→D3 VisA. the ensemble (Table 1) does not depend on these experts.

We benchmarked three expert models on the calibration split (Table 12). SubspaceAD dominates overall, reaching 1.000 AUROC on D11 VisA calibration where all VLMs score ≤ 0.85 . AnomalyVFM (zero-shot, DINOv2 + LoRA) is uniformly weaker (macro 0.599) and was excluded from the final system. DINOv2-PatchNN complements SubspaceAD on D03 infrastructure (0.80 vs 0.50). The oracle that selects the best expert per domain reaches macro 0.898, indicating 7 pp headroom for multi-expert routing as future work.

Table 12: Expert model comparison (calibration, macro AUROC).

Expert	Macro	Strongest domain
DINOv2-PatchNN	0.640	D09 road (1.00)
SubspaceAD	0.756	D11 VisA (1.00)
AnomalyVFM	0.599	D03 infra (0.72)
Oracle (max/domain)	0.898	—

E Descriptor ablation (legacy, taxonomy)

This ablation supports Finding 1 and uses the earlier D01–D11 display codes on the earlier twelve-domain split; the code mapping to the taxonomy is listed at the top of Appendix D. the ensemble’s Direct branch in Table 1 runs descriptor-free by design (§??), so the ablation is an orthogonal finding about prompt engineering on the Direct baseline rather than a component of the ensemble.

Table 13 shows the per-domain effect of replacing the generic descriptor (“Identify whether the query image is anomalous relative to the reference(s)”) with the task-anchored descriptor (Appendix A) on the test split for all three backbones. The macro gain is positive and statistically significant for every backbone (paired bootstrap with 1,000 resamples, per-domain stratification, $n=1,298$): GPT-5.4 +6.4 pp [+4.4, +8.4], SeedVL +4.1 pp [+1.6, +6.4], Qwen3.5 +3.2 pp [+1.0, +5.5]. The three backbones agree on the qualitative direction but disagree on which domains benefit most: GPT-5.4 is dominated by D06 Liver CT and D08 Change (+31/+33 pp), SeedVL gains most on D08 Change and D09 Road (+18/+14 pp), Qwen3.5 gains most on D06 Liver and D04 Dermoscopy (+14/+8 pp). A handful of small per-domain regressions appear on each backbone (e.g. D03 Infrastructure on all three, D09 Road on Qwen3.5); these are VLM-strong domains where the more specific prompt trims some true-positive confidence.

F Per-domain AUROC across three backbones (the ensemble)

Figure 4 shows per-domain AUROC for Direct, the refutation agent alone, and the ensemble across all twelve domains on each of the three backbones. Table 1 contains the exact macro numbers and bootstrap CIs; this figure visualises the per-domain dispersion and where the ensemble’s complementary-error advantage concentrates.

G Per-domain the specialty-aware variant vs the ensemble deltas

Table 15 lists per-domain AUROC for the the ensemble baseline and the the specialty-aware variant specialty-aware ensemble (§4.4) on the two backbones for which the the specialty-aware variant run completed cleanly (1418/1418 items, 0 errors). Both methods use the same parallel-Direct + the refutation agent ensemble at $\alpha=0.5$; only the tool catalog and agent prompt differ. ▲ marks domains where the specialty-aware variant improves by ≥ 1 pp; ▼ marks regressions of ≥ 1 pp.

Table 13: Descriptor ablation on the test split: per-domain AUROC (Generic vs Task-anchored, Δ in pp). Bold = largest per-domain gain per backbone.

Domain	GPT-5.4			SeedVL			Qwen3.5		
	Gen.	Task	Δ	Gen.	Task	Δ	Gen.	Task	Δ
D01 Industrial	0.951	0.962	+1.0	0.848	0.874	+2.6	0.912	0.903	-0.8
D02 Retail	0.752	0.774	+2.2	0.864	0.863	-0.1	0.632	0.672	+4.0
D03 Infrastructure	0.664	0.623	-4.1	0.745	0.660	-8.6	0.702	0.712	+1.0
D04 Dermoscopy	0.769	0.796	+2.7	0.708	0.760	+5.2	0.680	0.762	+8.2
D05 Brain MRI	0.915	0.934	+1.9	0.736	0.864	+12.8	0.866	0.849	-1.8
D06 Liver CT	0.433	0.745	+31.2	0.529	0.492	-3.7	0.544	0.684	+14.0
D07 GI Endoscopy	0.903	0.890	-1.3	0.881	0.876	-0.4	0.874	0.918	+4.4
D08 Change Det.	0.496	0.827	+33.2	0.643	0.822	+17.9	0.767	0.828	+6.1
D09 Road Obstacle	0.954	0.968	+1.4	0.793	0.936	+14.3	0.970	0.911	-6.0
D10 Logical	0.660	0.683	+2.3	0.678	0.651	-2.7	0.641	0.676	+3.5
D11 VisA	0.873	0.878	+0.5	0.805	0.878	+7.3	0.776	0.800	+2.4
Macro	0.761	0.825	+6.4	0.748	0.789	+4.1	0.760	0.792	+3.2
95% CI (Δ)	[+4.4, +8.4]			[+1.6, +6.4]			[+1.0, +5.5]		

Table 14: Per-backbone macro AUROC on CrossDomainVAD-12 test ($n=1,418$, 12 domains, ≥ 90 valid items per domain after error filtering): the ensemble’s Direct branch, the refutation agent alone, and the ensemble. Headline numbers match Table 1; full per-domain breakdown in Figure 4. All comparisons use the 1,000-resample stratified paired bootstrap with per-domain stratification.

Backbone	Direct	the refutation agent	Ens	$\Delta_{\text{Ens-D}}$	95% CI	$P(\Delta > 0)$	sig.
GPT-5.4	0.731	0.768	0.772	+4.01	[+2.58, +5.50]	1.000	✓
SeedVL	0.678	0.715	0.738	+5.99	[+4.32, +7.66]	1.000	✓
Qwen3.5	0.714	0.671	0.732	+1.69	[-0.08, +3.51]	0.971	(0)

The largest per-domain the specialty-aware variant gains are on Liver CT (+13.4 pp SeedVL, +12.9 pp Qwen3.5) and SDNET concrete (+6.9 SeedVL, +1.8 Qwen3.5)—both are domains where the new `tool_expert_score` either has strong native signal (D10 subspacead AUROC 0.89) or where `tool_texture_fft` can detect crack-induced periodicity disruption. Regressions cluster on two failure modes: (a) expert-unavailable industrial domains where the tool ladder over-engages on items the ensemble correctly resolved at turn 1 (D3 Qwen3.5 -5.6 pp); (b) balanced domains where the ensemble was already near-saturated and the extra variance from the specialty-aware variant’s added tool options is net-negative (D5 SeedVL -2.0 pp, D7 SeedVL -2.4 pp). Nothing in the specialty-aware variant uses dev information, so we do not tune against these regressions; the macro CI on both backbones excludes zero despite the per-domain dispersion.

H Leave-one-domain-out sensitivity (full table)

I Paired bootstrap (full nine-row table) and budget-matched controls

Table 5 in the main body lists the three GPT-5.4 rows; Table 17 below includes all nine pairwise comparisons across the three backbones. Table 18 reports the negative-control budget-matched baselines that we ran on GPT-5.4 and SeedVL calibration splits.

J VLM×Expert complementarity matrix (full figure)

K AnomalyClaw decision trace example

Every AnomalyClaw prediction carries a full trace that a human reviewer can audit. For illustration, we show two trace snippets from the actual test runs (Qwen3.5 backbone) first the autonomous router plan generated for one domain, then a per-item agent execution.

Autonomous router plan (D5c BMAD-Liver, Qwen3.5).

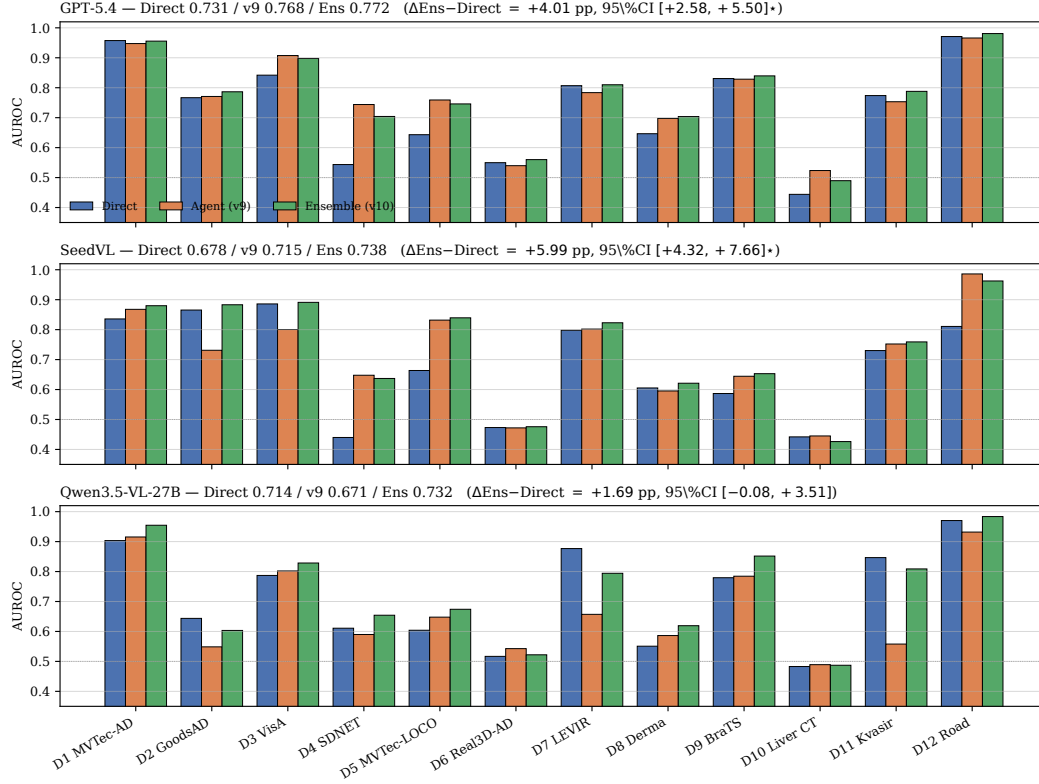


Figure 4: **Per-domain AUROC on CrossDomainVAD-12 test** ($n=1,418$), one panel per VLM backbone. Three bars per domain: Direct (blue) / refutation agent alone (orange) / ensemble 0.5 · Direct + 0.5 · the refutation agent (green). Grey dotted line marks the 0.5 random-guessing floor. Macro numbers and the stratified paired bootstrap $\Delta(\text{Ens}-\text{Direct})$ with 95% CI are reported in each panel title. The ensemble’s complementary-error gain concentrates on infrastructure (D4), logical (D5), and road (D12) on SeedVL; on GPT-5.4 the gain is uniform across industrial and medical families; on Qwen3.5-VL-27B the ensemble has to compensate for the refutation agent’s deficit on D7 (LEVIR) and D11 (Kvasir) by relying on Direct.

Strategy: interpret. **Expert:** subspacead. **Tools:** {domain_descriptor, hotspot_cropper}.

Reasoning (verbatim from Qwen3.5): “SubspaceAD is the designated expert for liver CT texture anomalies, and the interpret strategy provides a critical safety net to verify subtle focal lesions that the VLM may miss. The hotspot cropper makes the second-call evidence explicit when the expert flags a concentrated abnormality.”

Cost: 1 text-only VLM call (671 input / 1013 output tokens) amortised across all 120 D5c test items.

Per-item agent execution (D1 Industrial, item D1_0063, SeedVL).

Phase 1 (Plan lookup). Cached plan for D1 = (strategy:fusion, expert:subspacead, tools {domain_descriptor, hotspot_cropper}).

Phase 2 (Expert). SubspaceAD.score = 17.95 (global calibration median $m=40.0$); top-5 patches at (22, 24), (10, 24), (10, 22), (33, 32), (...) in the 48×48 grid. Derived $\rho = 159.7$, $\kappa = 1.43$.

Phase 3 (Online override). $\hat{y}=\text{normal}$ AND $\rho > 0.8 \Rightarrow$ promote strategy fusion $\rightarrow$ interpret.

Interpret call. VLM re-examines the cropped hotspot bounding box (image coords (307, 137)–(459, 478)). image_label=“anomalous”, confidence=0.85, evidence=“a linear scratch visible at the flagged region was missed at global inspection.”

Table 15: Per-domain specialty-aware minus baseline AUROC delta for the two backbones whose specialty-aware run completed ($n=1418$, 0 errors each). The macro and bootstrap-CI row matches Table 3. GPT-5.4 is omitted here; its specialty-aware run was blocked by sub2api upstream failure on $D5-D12$ and only 345 valid items remain (76% API-error rate).

Dom	Dataset	SeedVL			Qwen3.5-VL-27B		
		the ensemble	the specialty-aware variant	Δ pp	the ensemble	the specialty-aware variant	Δ pp
D1	MVTec-AD	0.880	0.891	+1.08▲	0.955	0.956	+0.12
D2	GoodsAD	0.883	0.878	−0.53	0.603	0.649	+4.56▲
D3	VisA	0.891	0.932	+4.06▲	0.829	0.772	−5.60▼
D4	SDNET	0.637	0.706	+6.91▲	0.654	0.672	+1.76▲
D5	MVTec-LOCO	0.839	0.819	−2.00▼	0.674	0.673	−0.07
D6	Real3D-AD	0.476	0.536	+5.99▲	0.522	0.517	−0.53
D7	LEVIR	0.823	0.799	−2.41▼	0.794	0.842	+4.79▲
D8	Derma	0.621	0.673	+5.15▲	0.619	0.630	+1.09▲
D9	BraTS	0.653	0.685	+3.24▲	0.852	0.841	−1.07▼
D10	Liver CT	0.426	0.560	+13.41▲	0.487	0.616	+12.91▲
D11	Kvasir	0.759	0.759	+0.00	0.809	0.817	+0.84
D12	Road	0.963	0.969	+0.65	0.984	0.990	+0.68
Macro		0.738	0.767	+2.90	0.732	0.748	+2.04
95% CI, $P(\Delta>0)$			[+1.01, +4.85], 1.000			[+0.05, +4.00], 0.981	

Table 16: Leave-one-domain-out macro AUROC gain of score fusion over . All 11 LOO splits are positive on every backbone.

Metric (pp)	GPT-5.4	SeedVL	Qwen3.5
Full Δ (all 11 domains)	+1.4	+1.9	+5.9
LOO min	+0.9	+1.3	+4.9
LOO max	+2.2	+3.1	+7.0
LOO std	0.33	0.45	0.55
Positive splits / 11	11 / 11	11 / 11	11 / 11

Final decision. anomaly; score = 0.85. Ground truth: anomalous. Agent escalated correctly via the online expert override.

Every test-split item has an analogous trace, serialised as JSON in the released `plan` and `raw_output` fields. Score fusion, by contrast, produces only a scalar.

L Route C: Component Enumeration prototype

We prototyped a fourth route, *Component Enumeration* (Route C), for logical anomalies (D10 MVTEC-LOCO) where the anomaly is defined by wrong count or wrong spatial arrangement of components rather than by a texture defect. Route C triggers when $\rho>1.2$ and $\kappa<1.10$ (strong but *dispersed* expert signal). The action is a second VLM call with a specialised enumerate prompt that asks the model to list components in the query versus the references and report count differences.

On D10 calibration (20 items, 4 reference images per query), Route C reaches 60% accuracy versus ’s 55%. The gain is modest for two reasons: (i) VLMs struggle to count small similar objects reliably, and (ii) the 20-item calibration split yields noisy threshold estimates. Route C is *not* part of the main-table AnomalyClaw variant reported in Table 1: the reported numbers use only Routes A/B/D. We include Route C here as a concrete direction for future work and flag it as a prototype rather than a validated component of the agent.

Table 17: Full stratified paired bootstrap table (1,000 resamples, per-domain stratification, $n=1298$).

Backbone	Comparison	Δ	95% CI	sig. at $p<0.05$
GPT-5.4	fusion vs	+0.014	[+0.003, +0.025]	✓
	agent vs	+0.005	[−0.007, +0.015]	—
	fusion vs agent	+0.009	[−0.004, +0.022]	—
SeedVL	fusion vs	+0.019	[+0.008, +0.031]	✓
	agent vs	+0.023	[+0.007, +0.039]	✓
	fusion vs agent	−0.005	[−0.019, +0.010]	—
Qwen3.5	fusion vs	+0.059	[+0.045, +0.076]	✓
	agent vs	+0.019	[+0.009, +0.030]	✓
	fusion vs agent	+0.040	[+0.026, +0.055]	✓

Table 18: Budget-matched negative controls (*/-era numbers, pre-*; included as historical record of disqualified designs and *not comparable* with the ensemble numbers in Table 1). TheDirect / SubspaceAD fusion / AnomalyClaw rows in this table use our older descriptor Direct and our router AnomalyClaw system; the main-paper headline numbers (GPT-5.4 0.864, SeedVL 0.809, Qwen3.5 0.814) are from the refutation agent+Direct ensemble on the same test split. “Random escalation” and “DINOv2 blend” are calibration-split numbers for disqualified designs. This table’s purpose is only to show that blind second-call escalation and weaker expert blends are negative controls; it should not be read as a competing test-split comparison against our headline recipe.

Method	Calls/img	GPT-5.4	SeedVL	Qwen3.5
Direct	1.00	0.822	0.795	0.792
Random escalation 30%	1.30	0.795	0.750	—
+ 0.2·DINOv2 blend	1.00	0.830	0.807	—
+ 0.2·SubspaceAD fusion	1.00	0.835	0.813	0.851
AnomalyClaw	~1.3	0.826	0.818	0.811

M Routing distribution

N Proposer–Advocate debate protocol

The *debate* strategy runs two VLM calls in sequence. The Proposer receives the query, references, and the task-anchored descriptor, and returns a JSON object {image_label, anomaly_type, evidence, confidence, bbox}. The Advocate receives the Proposer’s output verbatim plus an adversarial prompt asking it to refute the Proposer’s claim using the same references, and returns a JSON object {refute_confidence, counter_evidence}. We combine both into a final score via the rule: if `refute_confidence` ≥ 0.6 , overturn the Proposer; if `proposer.confidence` ≥ 0.6 and `refute_confidence` ≤ 0.4 , commit to the Proposer; otherwise take the Proposer’s score unchanged. No third VLM call and no oracle. Temperature is 0 for both calls; the same descriptor ϕ_d is used.

O Descriptor router rules

The descriptor router maps each domain family to a default strategy. Rules were frozen once from the benchmark descriptor before any test query; no calibration AUROC was consulted.

P Tool contracts

Each tool has a small Python contract. Below are the full signatures; the implementations (Python/NumPy) are in `refine-logs/anomaclaw_v2/registry.py`.

- `domain_descriptor(domain_code) → str`: returns the task-anchored descriptor. Zero cost, no side effect.

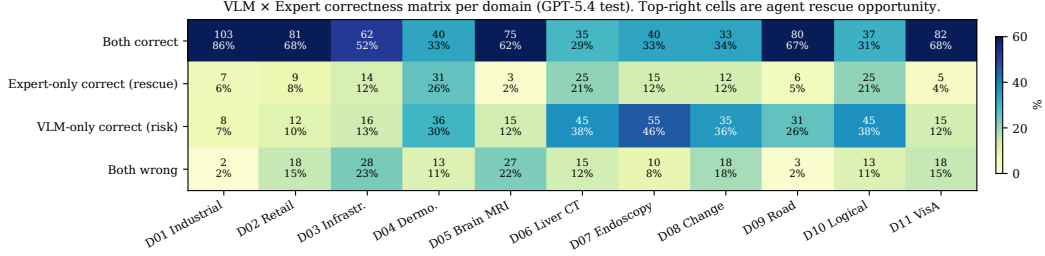


Figure 5: VLM×Expert correctness matrix per domain (GPT-5.4 test). Numbers are % of items. “Expert-only correct” is the upper bound of what the agent can recover; “VLM-only correct” is what it risks losing by blindly trusting the expert.

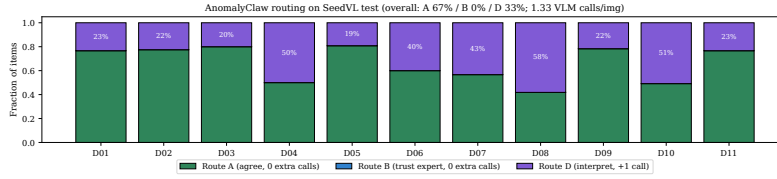


Figure 6: AnomalyClaw routing distribution on the SeedVL test split. 67% of items resolve by Route A (VLM and expert agree, 0 extra calls); Route D (interpret) is invoked on 33% of items; Route B (trust expert) does not fire on any test item because the asymmetric anomaly-trust gate handles strong concentrated signals before they reach Route B.

- `reference_retriever(query, refs, k=3) → list[int]`: cosine similarity of DINOv2-CLS tokens; returns top- k reference indices.
- `hotspot_cropper(query, patches, k=5, pad=0.15) → bbox`: takes the top- k expert patches, computes a padded bounding box in image coordinates, returns a crop for the VLM.
- `component_counter(hotspot_map, threshold=0.5) → (n, stats)`: connected-component labelling with skimage; returns component count and per-component area/centroid.
- `knowledge_lookup(domain_code) → list[str]`: returns a fixed, per-domain keyword list (length ≤ 6 ; written once during benchmark setup).

Tools are composed explicitly by each strategy: *direct* uses only the descriptor; *fusion* uses descriptor and expert scores; *interpret* additionally uses the cropper and expert hotspots; *debate* uses the descriptor and optionally the retriever. No tool is invoked outside its strategy’s declared contract.

Q Rulebook Pilot: L1 invariants + L2 oracle corrective rules

The domain-rule corpus reused by the Controller (§5, Stage D) was produced by an earlier pilot that we ran on the same CrossDomainVAD-12 dev/test split. We document it here for reproducibility; the pilot’s standalone results are not part of the main claim but its artefacts (per-domain JSON rulebooks) are the `invariant`, `corrective_fn`, and `corrective_fp` entries that the the controller extension rulebook RAG retrieves.

L1 — Reference-only invariants. For each (domain, category) pair, a reflector VLM (Qwen3.5-VL-72B) is shown $n=8$ normal reference images and asked to list visual *invariants* — concrete properties verifiable in *every* reference — restricted to six types:

- `count`: exact number of parts, segments, or discrete items.
- `symmetry`: axes or symmetries preserved across references.
- `spatial_layout`: global arrangement of components.
- `color_palette`: colour range observed across refs.

Table 19: Descriptor router rules.

Family	Strategy	Reasoning
industrial / retail / industrial_visa	fusion	texture-dominant few-shot; expert is reliably discriminative, VLM calibrates
infrastructure	fusion	concrete cracks flagged by patch expert; descriptor narrows false positives
logical (D9 MVTec-LOCO)	fusion	component structure visible to patch expert; VLM rules out semantic look-alikes
dermoscopy (D4)	fusion	patch expert catches localised lesions; VLM rules out imaging artefacts
medical_mixed / liver_ct / brain_mri	fusion	SubspaceAD complements VLM on non-trivial intensity shifts
gi_endoscopy (D7, D5d)	direct	VLM-strong: mucosa colour/shape semantics beyond generic feature distance
remote-sensing change (D6 LEVIR)	direct	change semantics require descriptor-level task anchoring
road (D8, D9 RA21+BDD)	direct	semantic obstacle recognition dominates; expert is a distractor

- **texture:** surface or fabric texture properties.
- **structural:** overall shape, topology, or part connectivity.

The reflector is forbidden from listing “hypothesised anomaly modes” (an earlier pilot design that collapsed into generic VLM-prior recitation). Empty lists are allowed when no invariant can be verified across all references.

L2 — Oracle cluster rules. For each domain, we run Passive the refutation agent on the 40-item dev split, form a balanced batch of $K/2$ false negatives ($gt=1, s<0.5$) plus $K/2$ false positives ($gt=0, s\geq 0.5$) with $K=10$, reveal their manifest labels, and ask a reflector VLM to propose 1–3 *corrective* rules per side over the whole batch at once. Per-item rules are forbidden (they over-generate); *normal_tolerance* rules must name a specific visual feature rather than blanket-suppress an anomaly class.

Rule store. For each domain we emit one JSON file containing the invariants and correctives tagged with metadata: (type, category [or null], text, source_items, confidence). The Controller never consumes these rules directly; instead it reads RAG-retrieved top- k subsets filtered by the query’s (domain, category) metadata (§5.1). The stacked store size ranges from 10 rules (D12 road safety, single category with few invariants) to 86 rules (D1 MVTec-AD, eight categories each contributing count/colour/texture/structural invariants).

Artefacts. Per-domain JSON: `benchmark/results/verbalized/_11/` and `_12/`. The the controller extension stacked rulebook: `benchmark/results/verbalized/_rulebook/`.